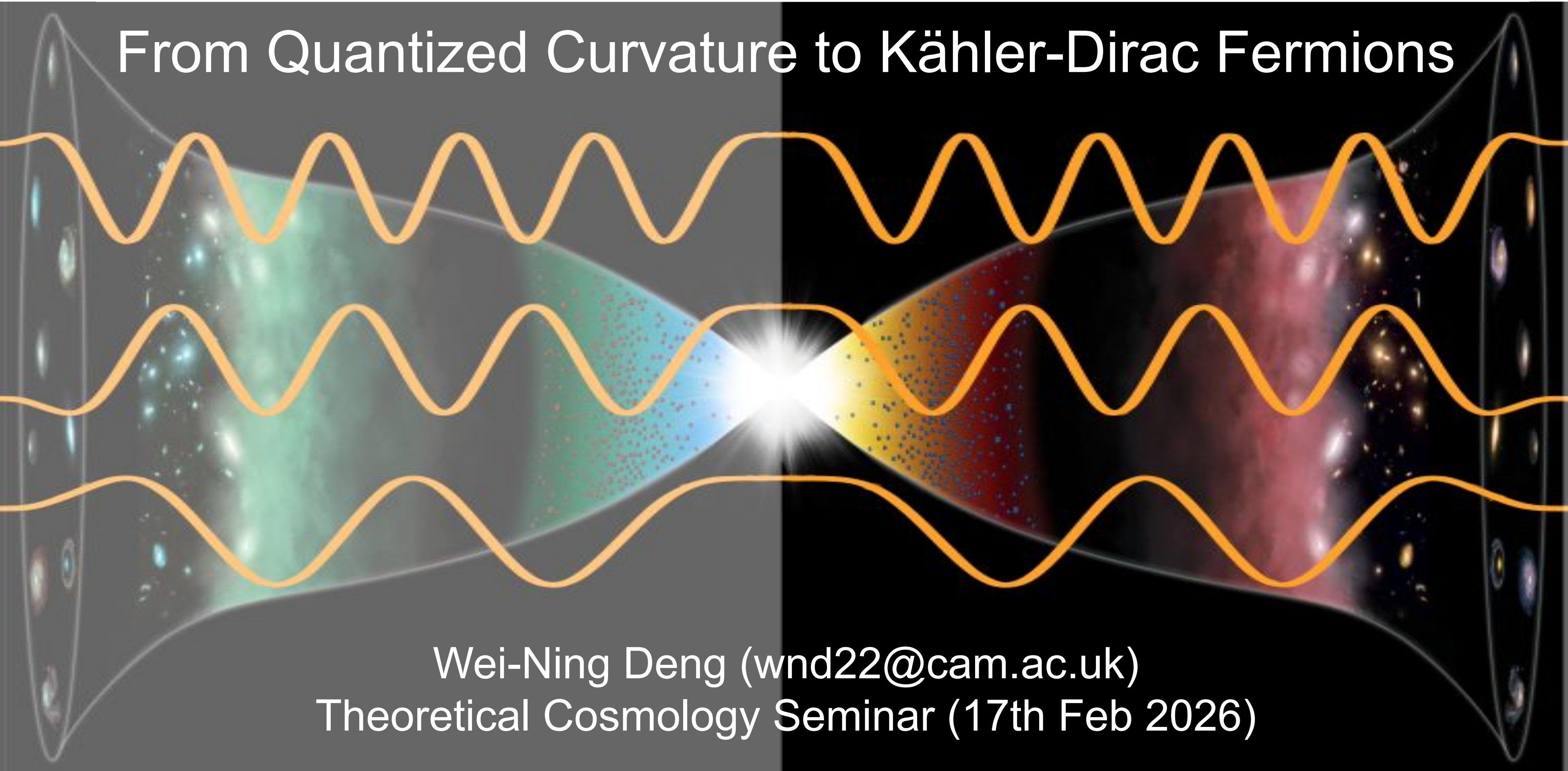
The CPT-Symmetric Universe:

From Quantized Curvature to Kähler-Dirac Fermions



Wei-Ning Deng (wnd22@cam.ac.uk)

Theoretical Cosmology Seminar (17th Feb 2026)

Outline

Part 1: Introduction – CPT-symmetric universe

Part 2: Theoretical Prediction of spatial curvature (PRD.110.103528, PRD.113.023546)

Part 3: Kähler-Dirac Fermions – Physical explanation for CPT (arxiv 2511.11548)

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Part 1: Introduction – CPT-symmetric universe

Part 2: Theoretical Prediction of spatial curvature

Part 3: Kähler-Dirac Fermions – Physical explanation for CPT

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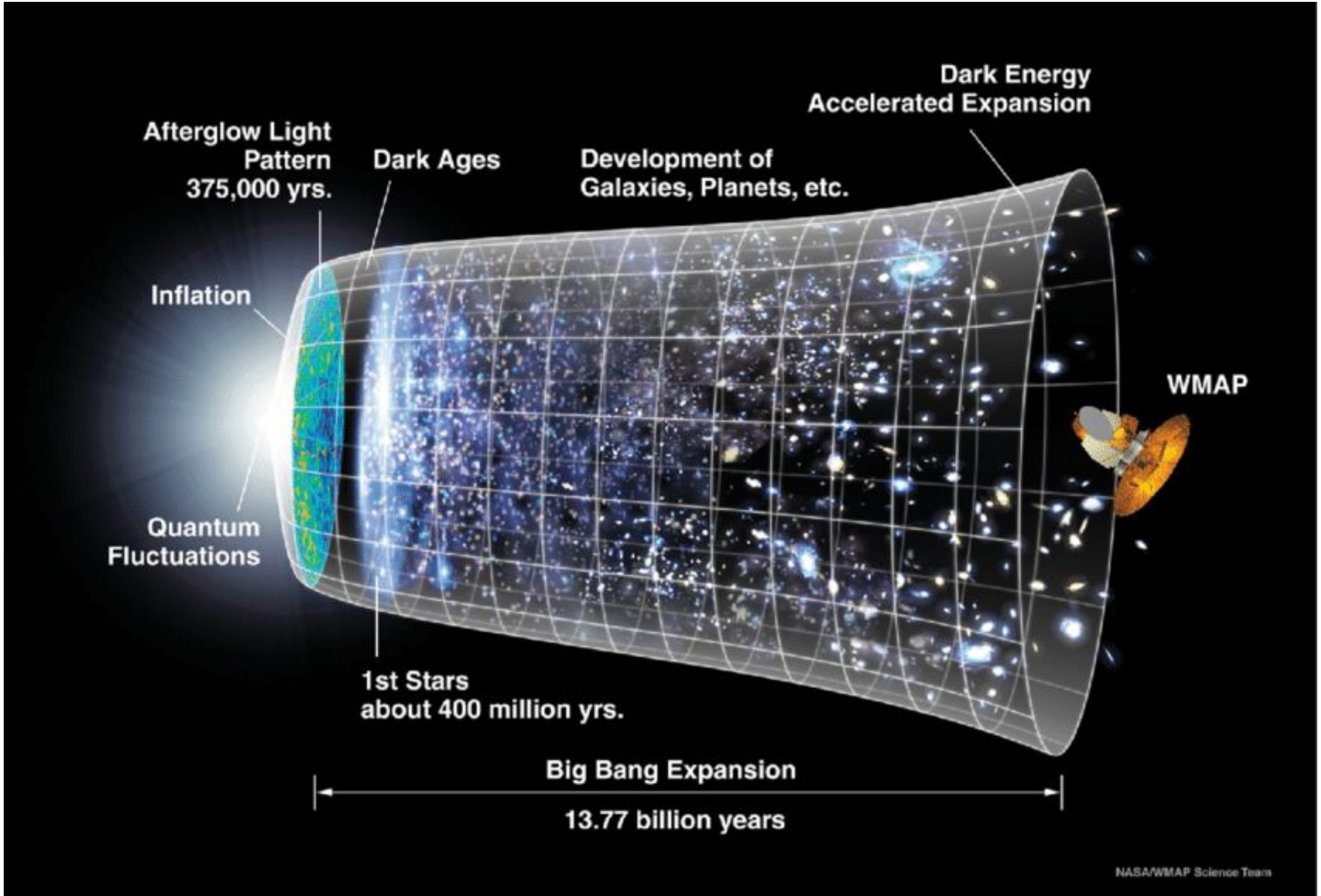
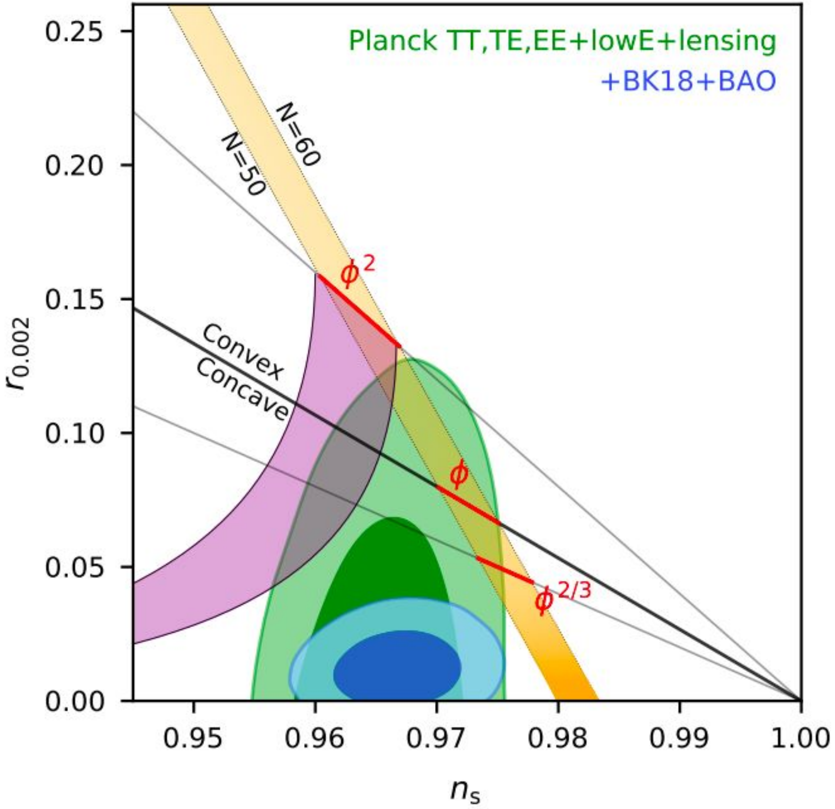
LCDM + SM PP + cosmic inflation



Theoretical and observational challenges

1. Problems in cosmology: Cosmological constant problem, Nature of DM, etc
2. Problems in PP: Hierarchy problem, strong CP problem, etc.
3. Observational challenges: BICEP/Keck [2203.16556] shows tider constraint on Inflationary B-modes.



LCDM provides a remarkably simple description of the large-scale universe: just 5 fundamental physics parameters

the matter/energy content

1. ρ_Λ cosmological constant
2. ρ_{DM}/ρ_B DM/baryon density
3. n_B/n_γ baryons per photon

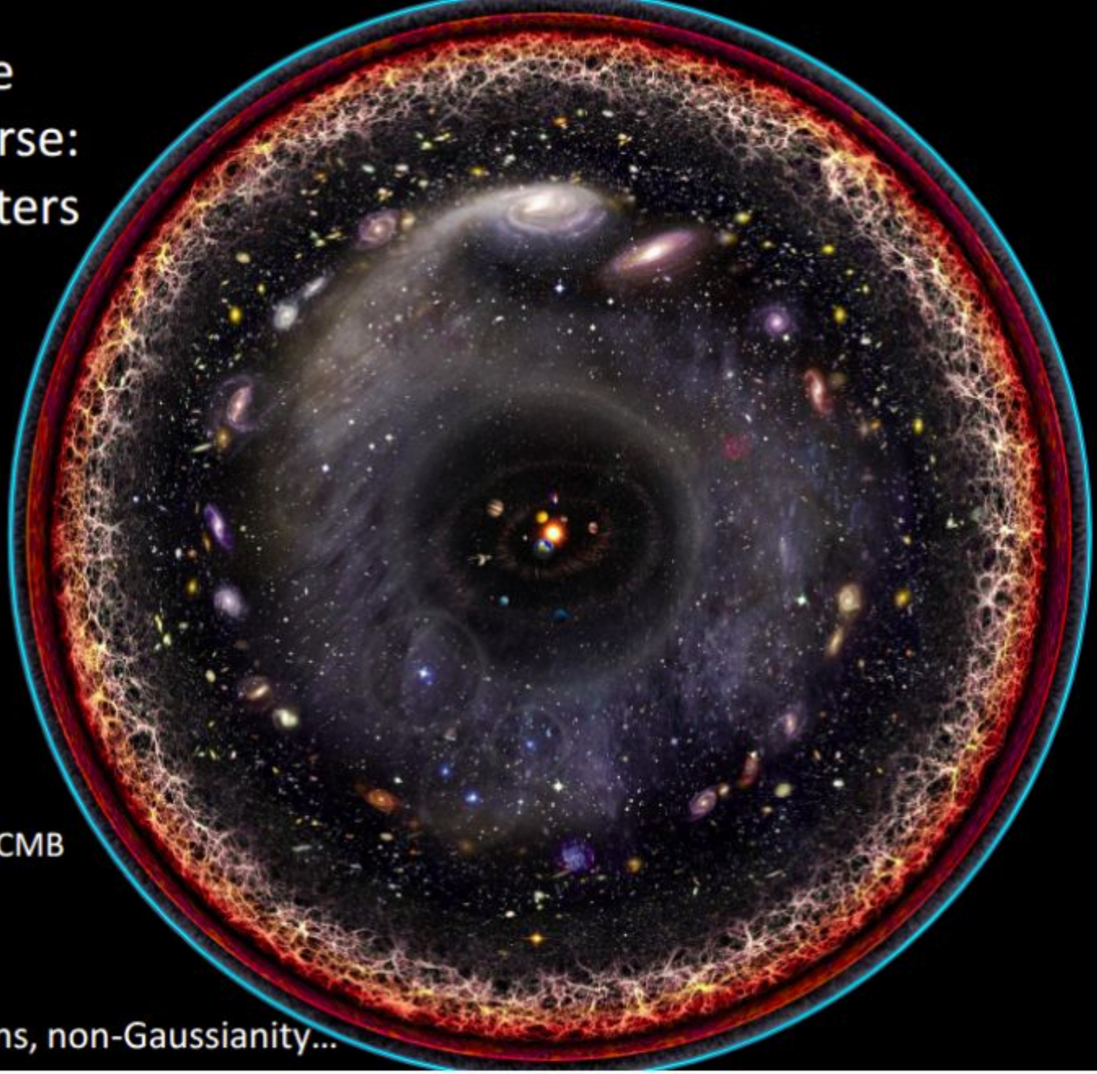
the large-scale geometry

Large scale Newtonian potential

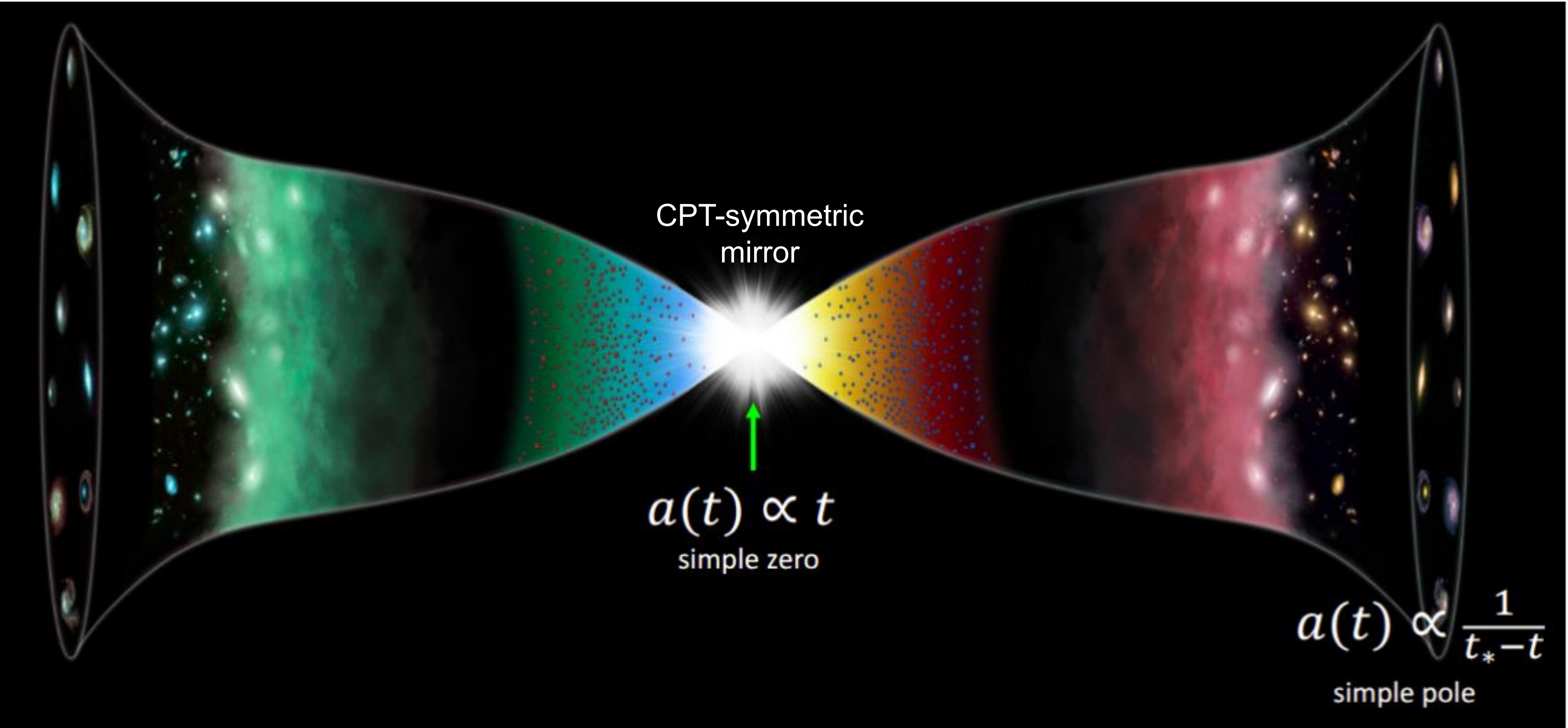
$$\langle \Phi^2 \rangle = \int \frac{dk}{k} A_\Phi \left(\frac{k}{k_*}\right)^{n_s - 1} \quad (k_* \equiv 0.05 \mathrm{Mpc}^{-1})$$

$$\left.\begin{array}{l} 4.\ A_\Phi = (7.6 \pm 0.1) \times 10^{-10} \\ 5.\ n_s - 1 = -0.04 \pm .006 \end{array}\right\} \text{Planck CMB}$$

many quantities are so far consistent with zero:
space curvature, tensor/isocurvature perturbations, non-Gaussianity...

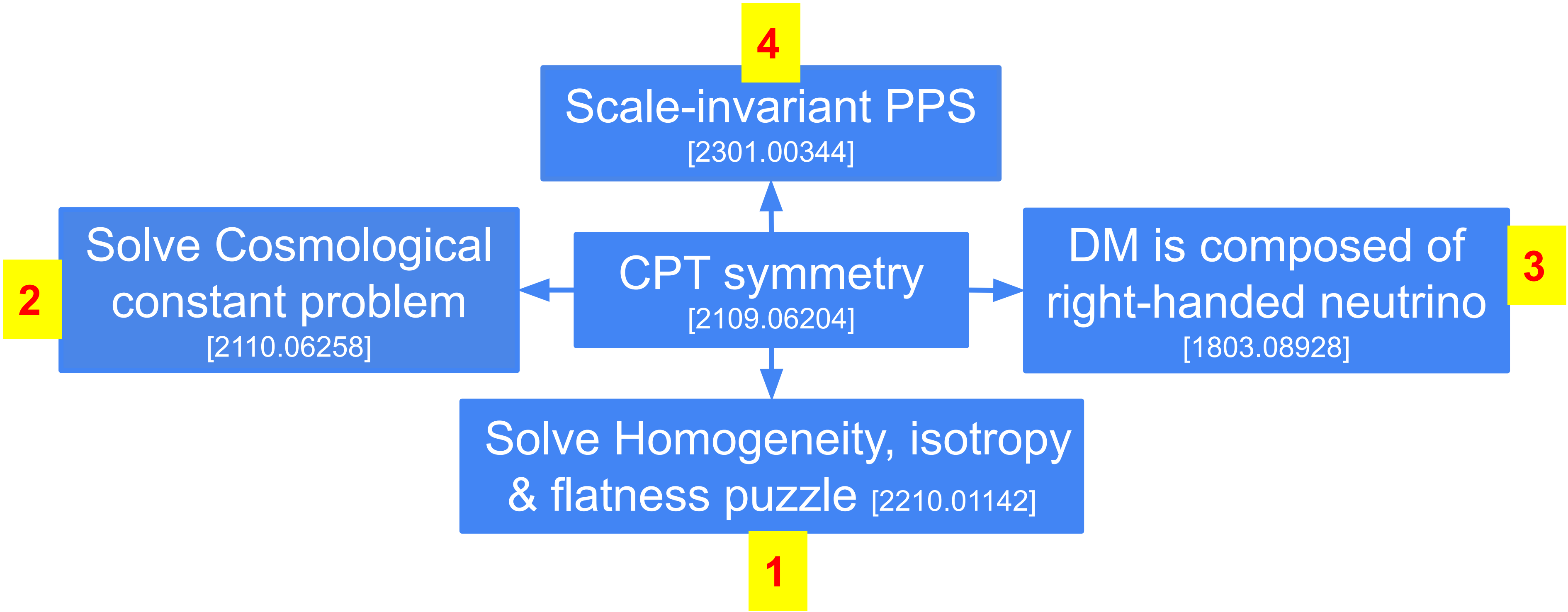


CPT-symmetric universe



CPT-symmetry universe can solve lots of problems

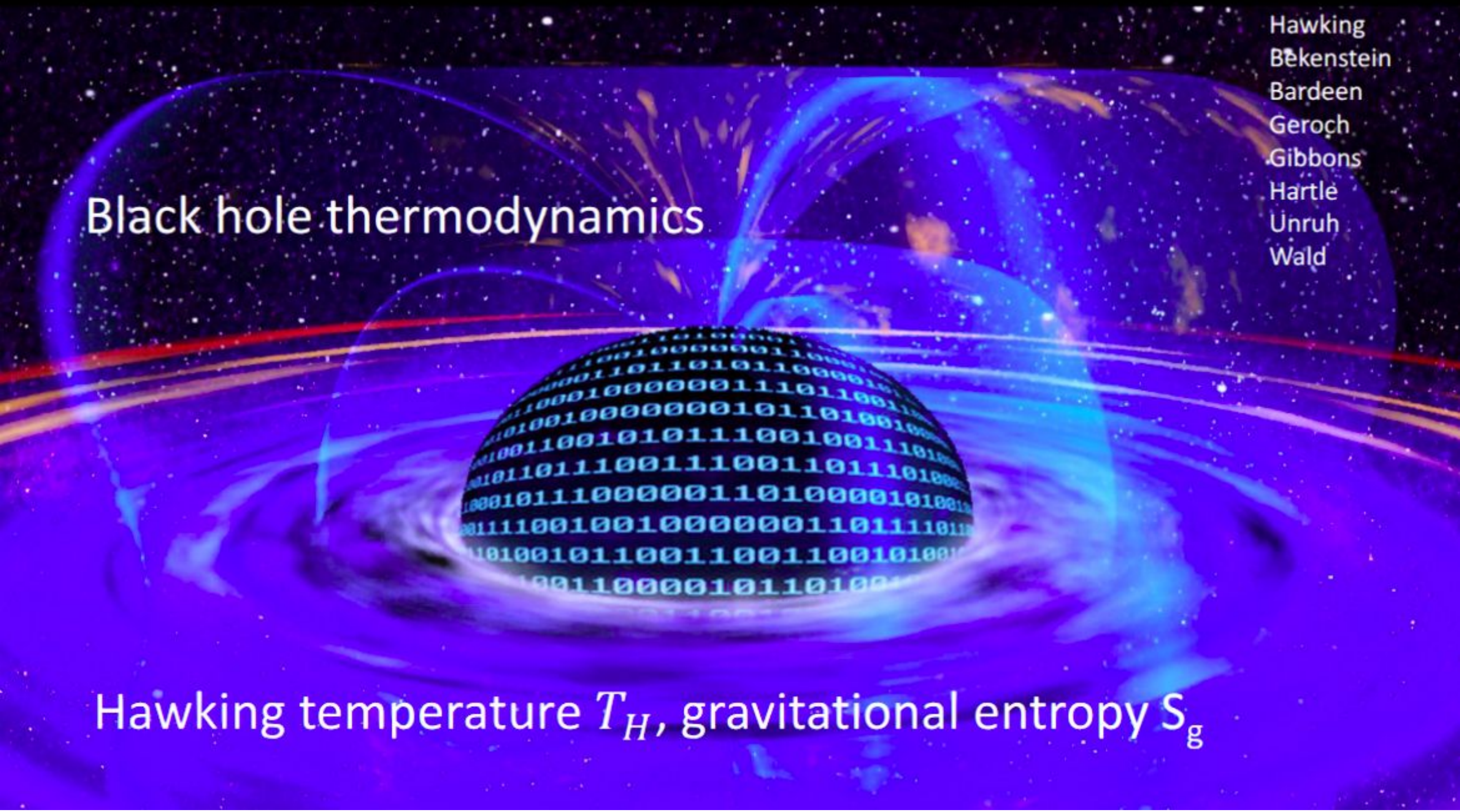
(Proposed by Latham Boyle and Neil Turok)



Inflation

10^{-32} seconds

10^{26} times larger


Hawking
Bekenstein
Bardeen
Geroch
Gibbons
Hartle
Unruh
Wald
Black hole thermodynamics
Hawking temperature T_H, gravitational entropy S_g

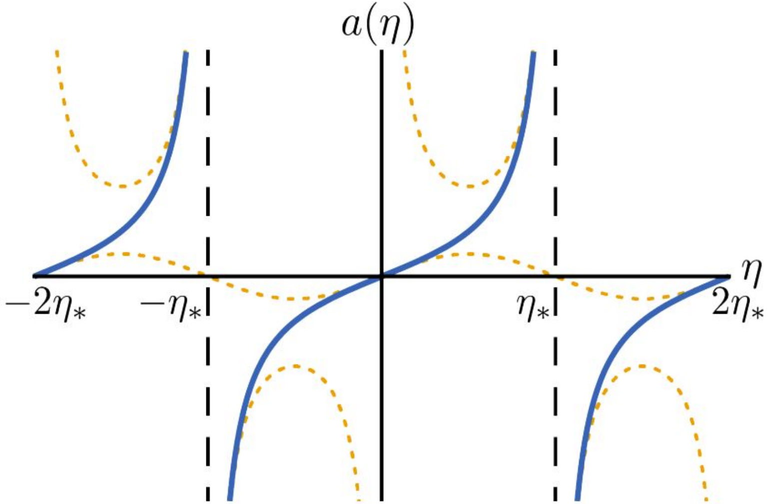
Analytic solution of Friedmann equation in comoving time

scale factor

$$ds^2 = a(t)^2\left(-dt^2 + \gamma_{ij}dx^i dx^j\right)$$

conformal time

comoving symmetric space (assume compact)



In Planck units

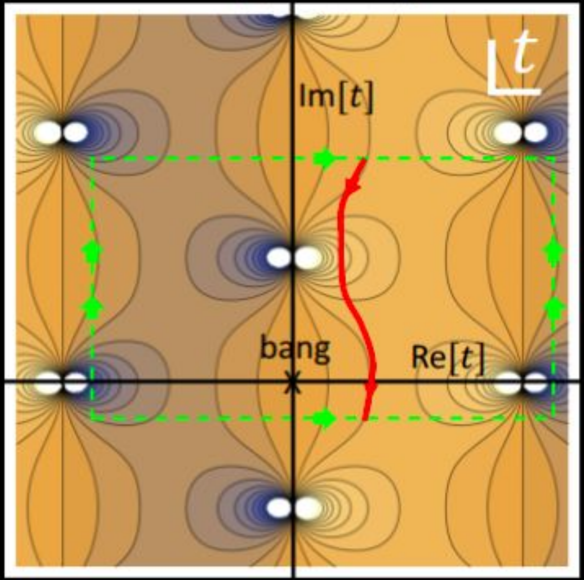
radiation matter space curvature Lambda

Friedmann $$3\dot{a}^2 = r + \mu a - 3\kappa a^2 + \lambda a^4$$

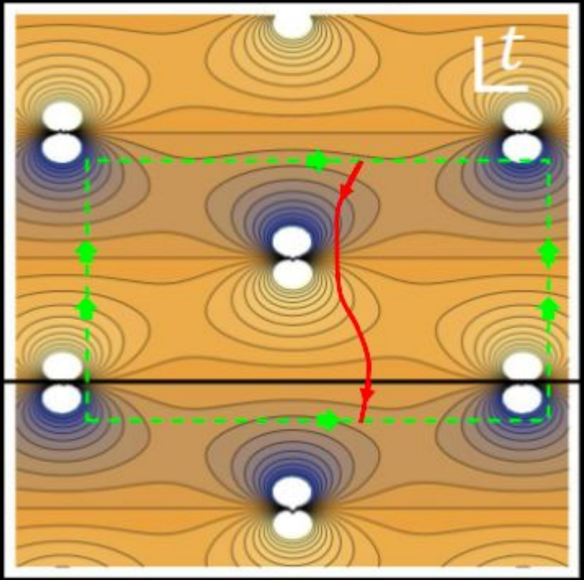
general solution (Jacobi elliptic function) is periodic in imaginary time

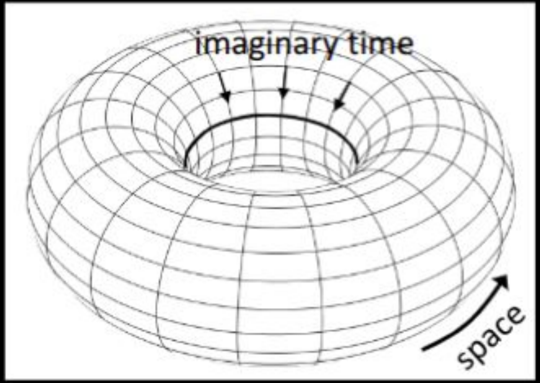
$a(t)$ is single-valued and doubly periodic in the complex t-plane: its only singularities are simple poles. The imaginary time period and the action computed over a period determine T_H and the gravitational entropy S_g

Re$[a(t)]$



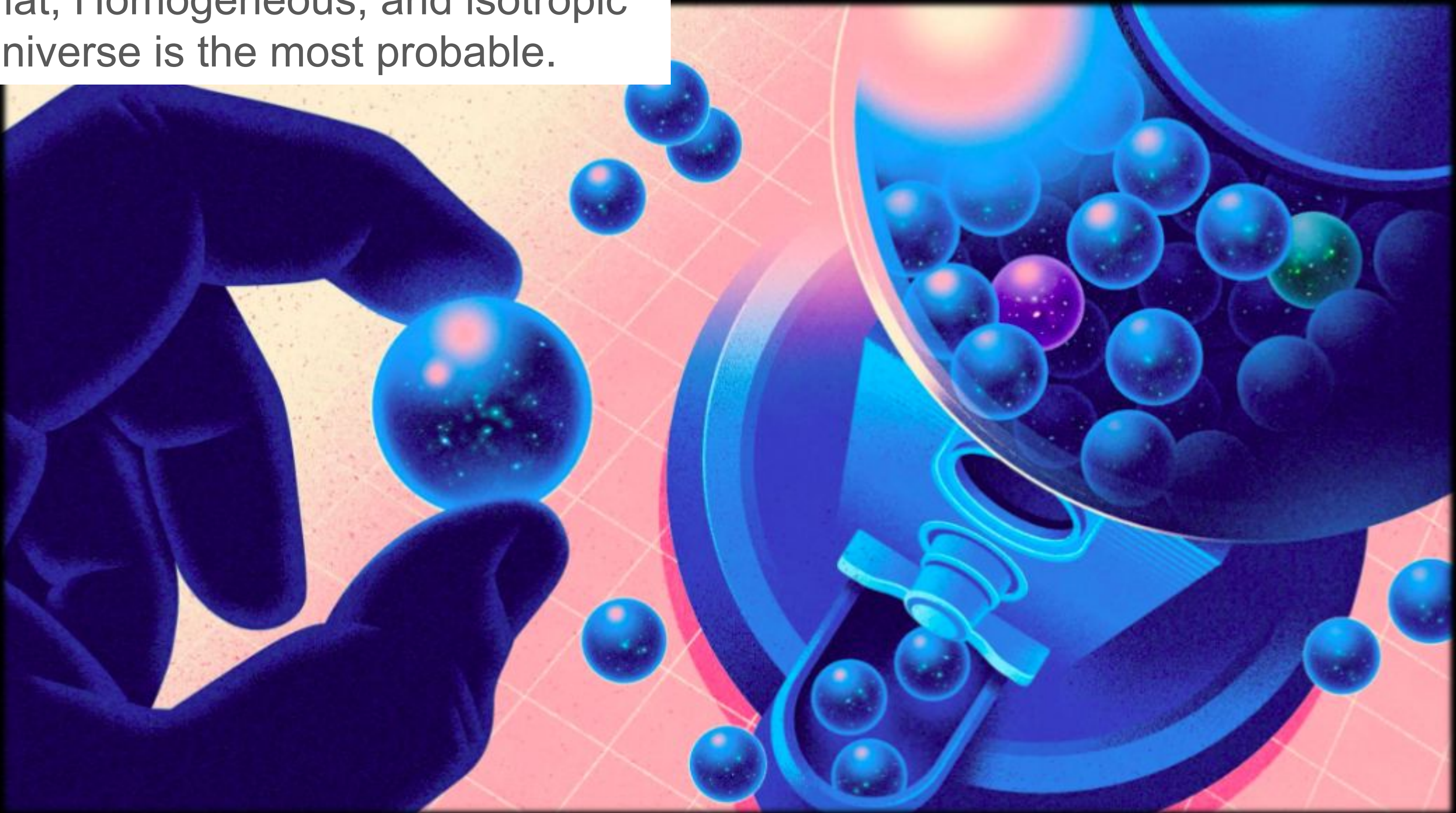
Im$[a(t)]$





Euclidean instanton for a universe w/radiation, matter, curvature, Lambda

Flat, Homogeneous, and isotropic universe is the most probable.



Quanta Magazine, Nov 17, 2022; *WIRED*, Jan 22, 2023

These fields can cancel the above 3 anomalies in coupling the SM to gravity

L. Boyle+NT, arXiv:2110.06258

Dim 1 scalars, fermions, gauge bosons, Dim 0 scalars

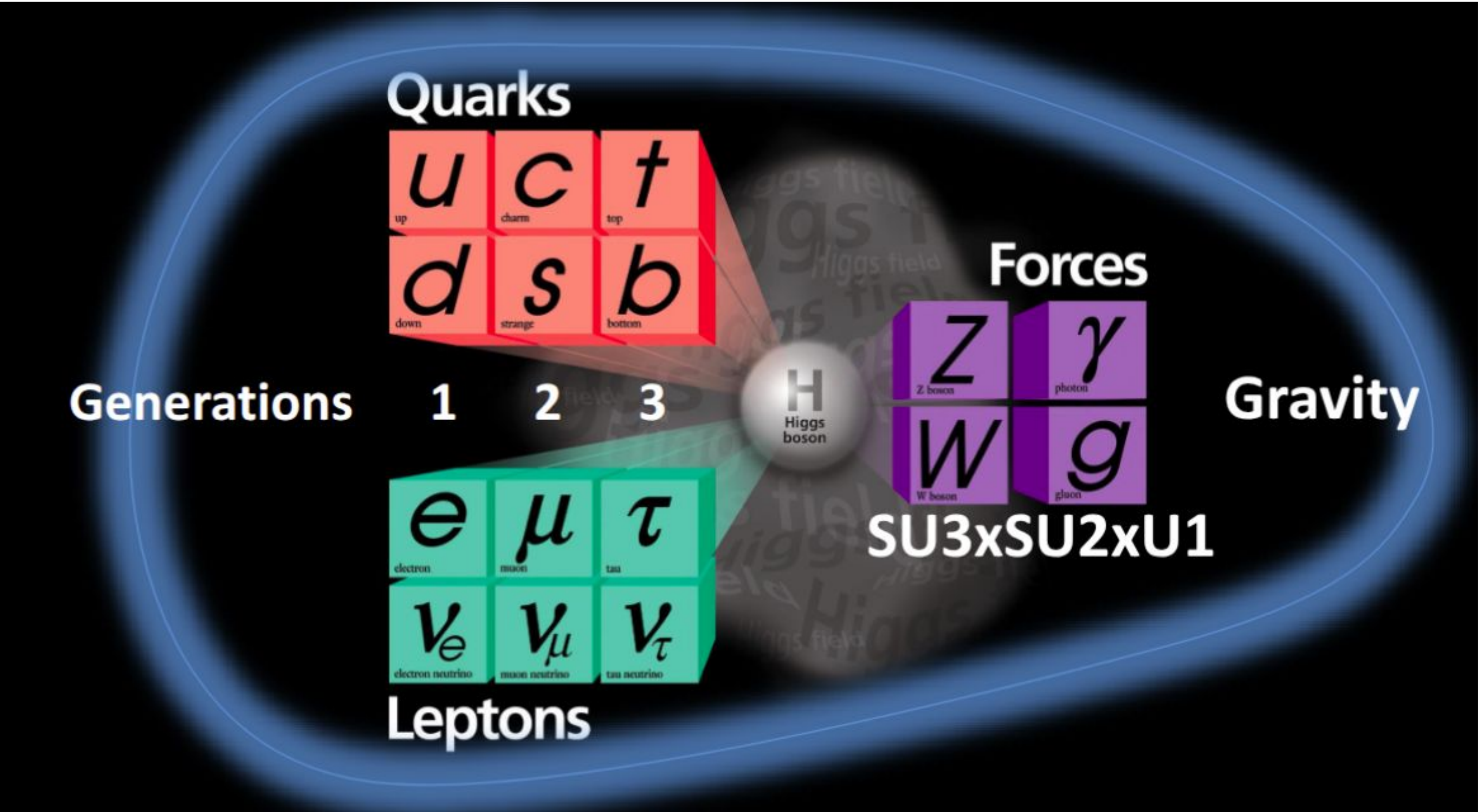
1. Vacuum energy $\propto n_{s,1} - 2n_F + 2n_A + 2n_{s,0}$
2. Conformal anomaly (Euler) $\propto n_{s,1} + \frac{11}{2} n_F + 62\, n_A - 28\, n_{s,0}$
3. Conformal anomaly (Weyl^2) $\propto n_{s,1} + 3\, n_F + 12\, n_A - 8\, n_{s,0}$

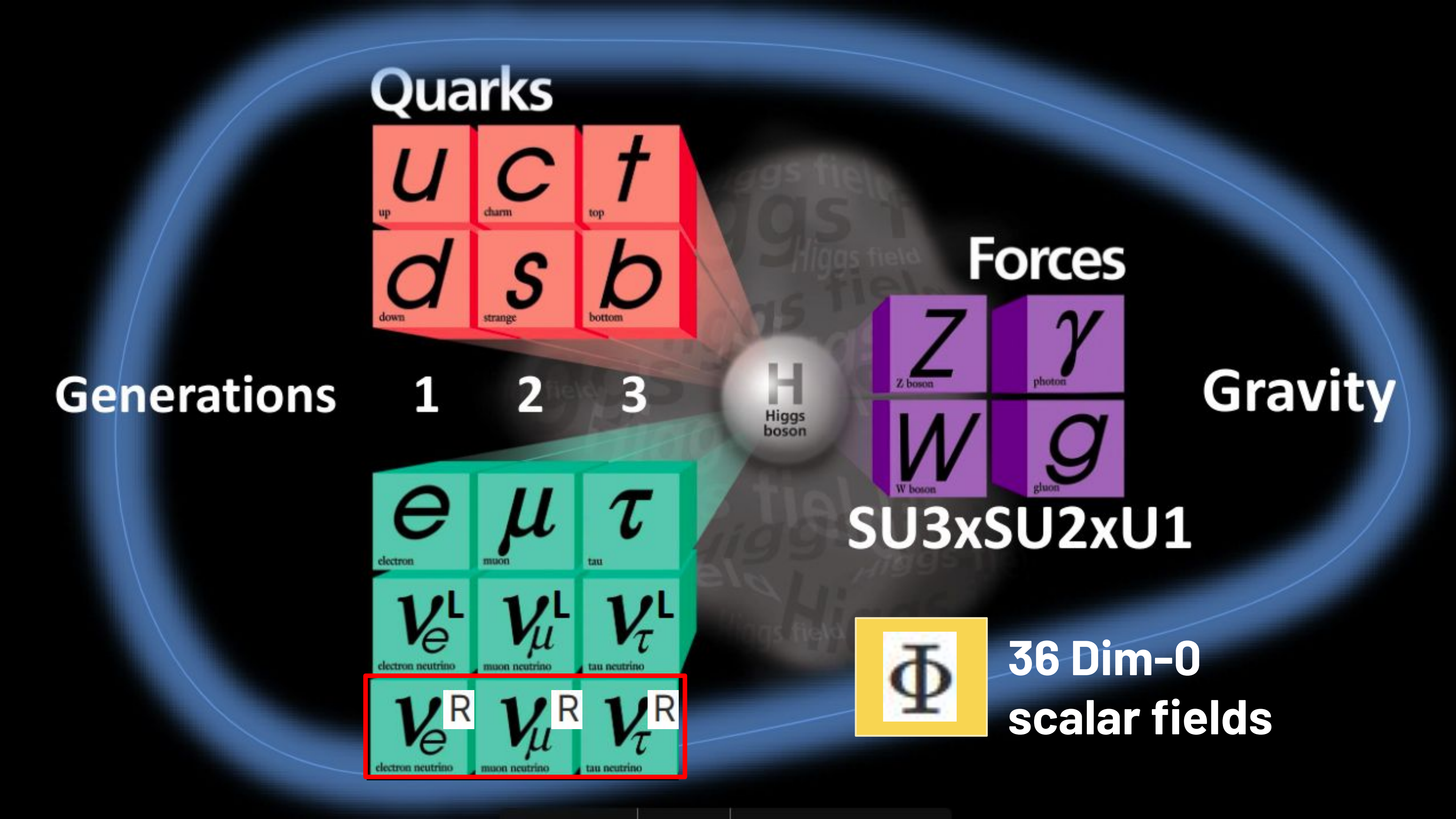
1) All three vanish iff $n_{s,1} = 0 \Longrightarrow$ no fundamental dimension one scalars (the Higgs must be composite)

2) Any two equations then give $n_F = 4n_A$ and $n_{s,0} = 3n_A$

3) For gauge group $SU3 \times SU2 \times U1$, *predict* n_F =48, *i.e.*, 3 fermion generations, each with a RH ν

Cosmological constant problem




Quarks
u
up
c
charm
t
top
d
down
s
strange
b
bottom
Generations
1
2
3
H
Higgs boson
Forces
Z
Z boson
γ
photon
W
W boson
g
gluon
Gravity
e
electron
μ
muon
τ
tau
ν_e^L
electron neutrino
ν_μ^L
muon neutrino
ν_τ^L
tau neutrino
ν_e^R
electron neutrino
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muon neutrino
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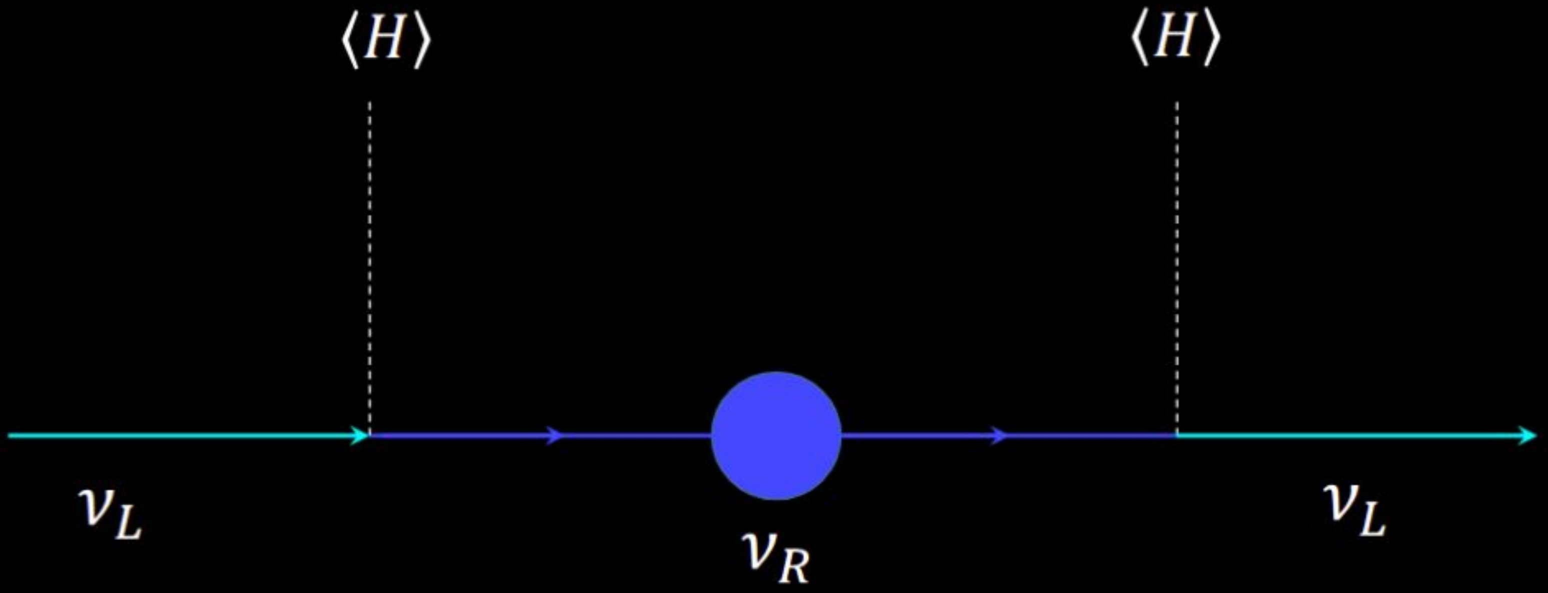
tau neutrino
SU3xSU2xU1
Φ
36 Dim-0
scalar fields

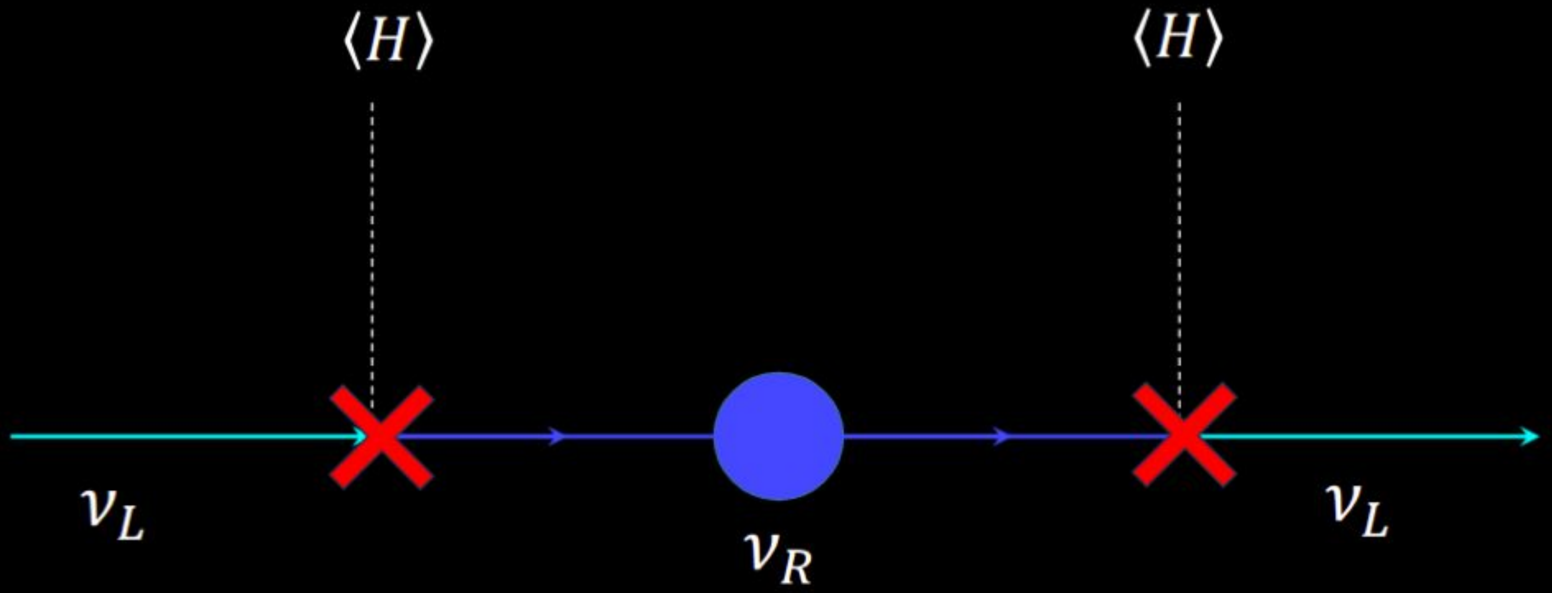
Right-handed neutrinos:



explain observed light neutrino masses
(seesaw mechanism, 1970's)

Later, I'll mention a new explanation for why ν_R's must exist

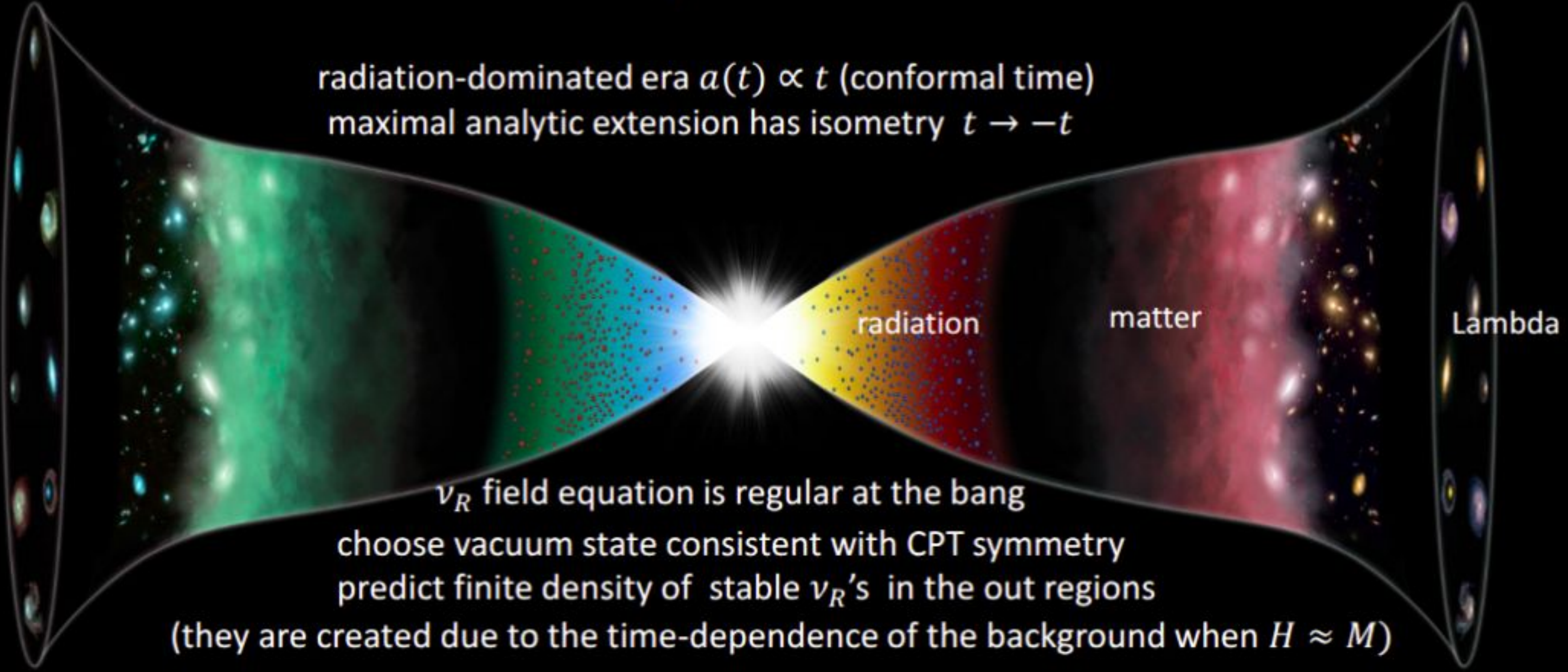
Stability of one RH neutrino $\Rightarrow$ $\mathbb{Z}_2$ symm $\Rightarrow$ lightest ν massless



will be tested using EUCLID, LSST and S4

a right-handed neutrino as the dark matter in a CPT-symmetric universe

radiation-dominated era $a(t) \propto t$ (conformal time)
maximal analytic extension has isometry $t \rightarrow -t$



ν_R field equation is regular at the bang
choose vacuum state consistent with CPT symmetry
predict finite density of stable ν_R's in the out regions
(they are created due to the time-dependence of the background when $H \approx M$)

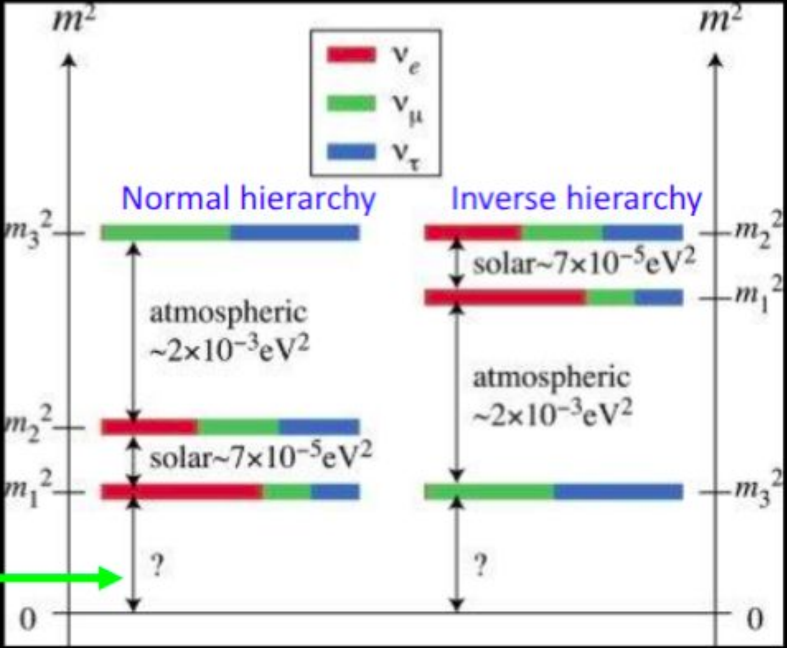
if one ν_R is stable, its density matches Ω_{DM} if its mass $M \approx 5 \times 10^8 GeV$

Light neutrinos: observations

current data

eBOSS 2007.08991

we predict zero



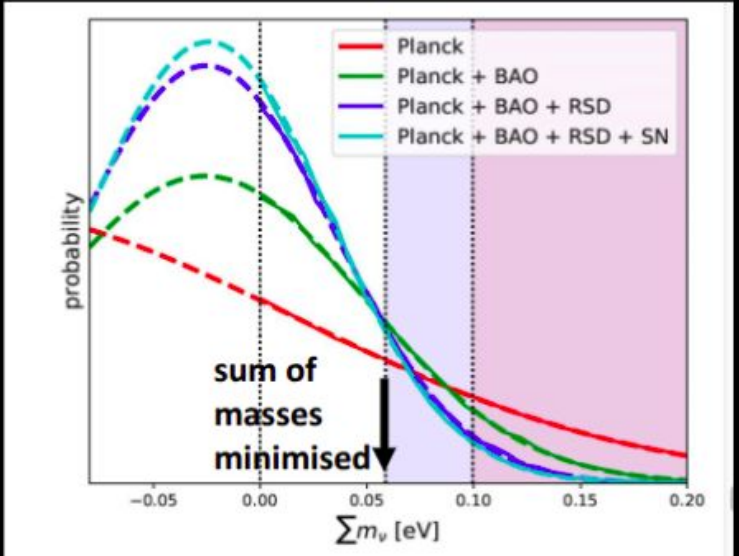
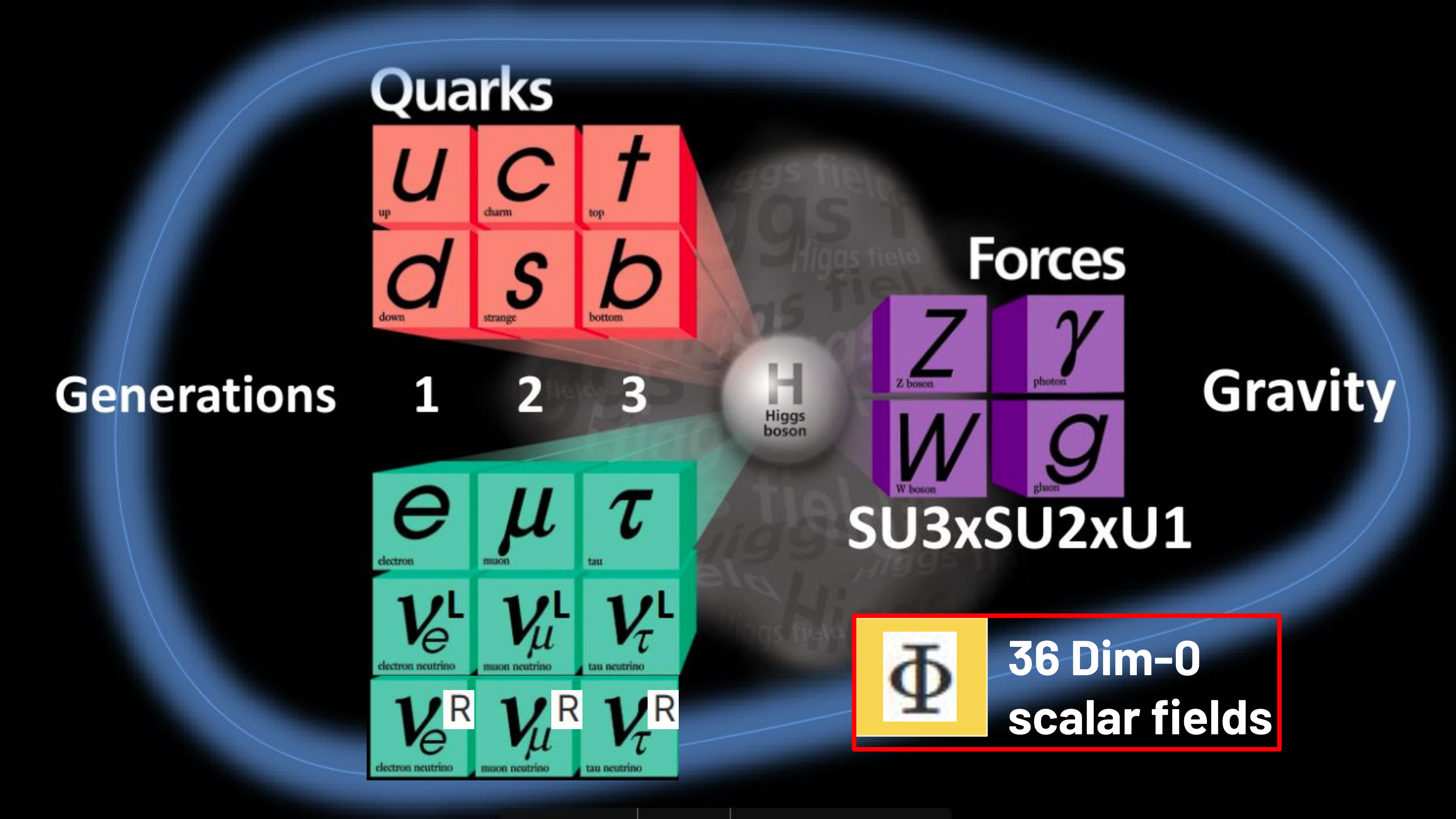


FIG. 13.— Posterior for sum of neutrino masses for selected con binations of data with a $\nu\Lambda$CDM cosmology. Dashed curves sho the implied Gaussian fits. Shaded regions correspond to lower lin its on normal and inverted hiearchies. Likelihood curves are no malized to have the same area under the curve for $\sum m_\nu > 0$.

Normal hierarchy: $M_\nu \equiv \sum m_\nu \approx 0.06\ eV$

Inverted hierarchy: $M_\nu \approx 0.1\ eV$


Quarks
u
up
c
charm
t
top
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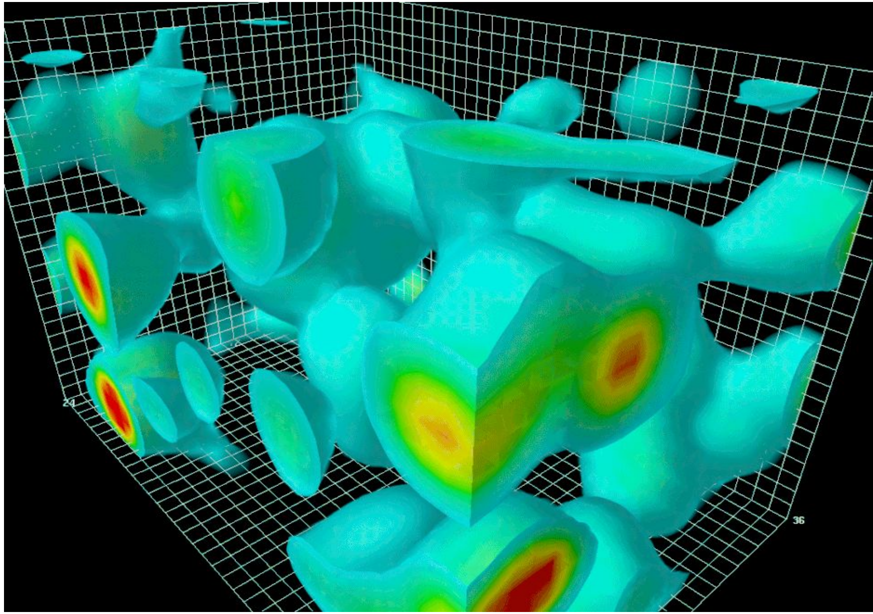
down
s
strange
b
bottom
Generations
1
2
3
e
electron
μ
muon
τ
tau
ν_e L
electron neutrino
ν_μ L
muon neutrino
ν_τ L
tau neutrino
ν_e R
electron neutrino
ν_μ R
muon neutrino
ν_τ R
tau neutrino
H
Higgs boson
Forces
Z
Z boson
γ
photon
W
W boson
g
gluon
SU3xSU2xU1
Gravity
Φ
36 Dim-0
scalar fields

Dim-0 scalar field seeds scale-invariant PPS

S = $-\frac{1}{2}\int d^4x\sqrt{-g}\,\varphi\Delta_4\varphi$

$$\langle\delta\varphi_i(\eta,\mathbf{x})\delta\varphi_j(\eta,\mathbf{x}')\rangle = \delta_{ij}\int\frac{d^3\mathbf{k}}{(2\pi)^3}\frac{\mathrm{e}^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}}{4ak^3},$$

$$\mathcal{P}_{\mathcal{R}} = \frac{3^2 5^3}{7(2\pi)^4}\left(\frac{c_\beta^{SM}}{\mathcal{N}_{eff}}\right)^2$$

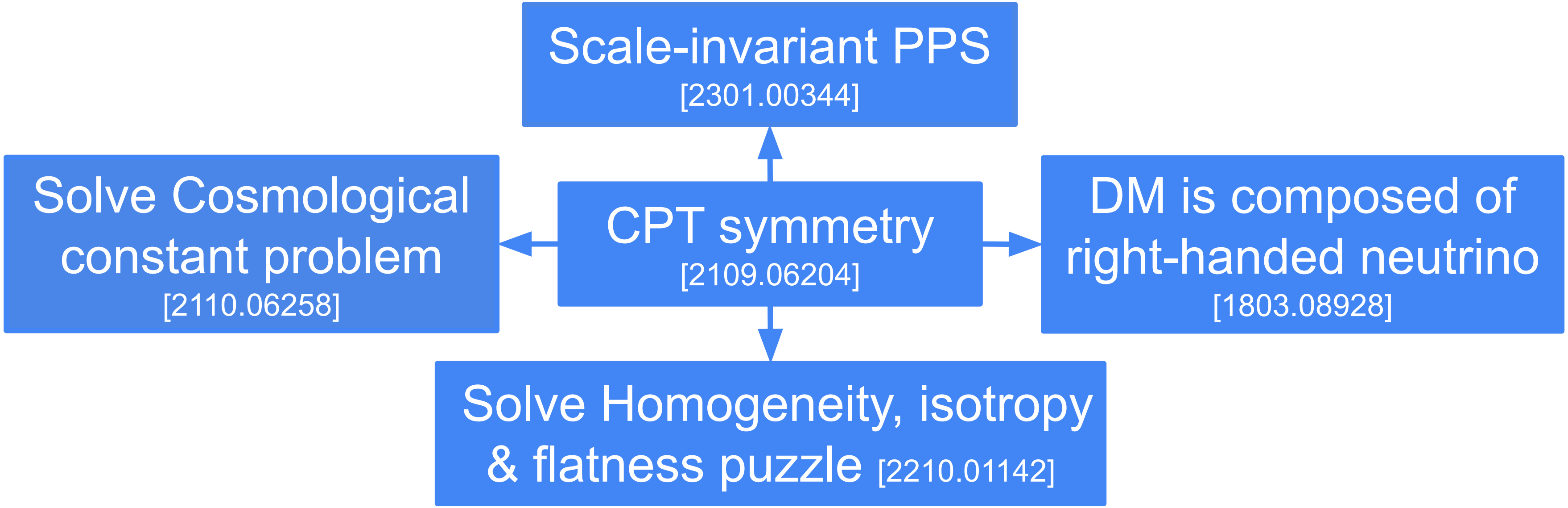


Predicted: $A = 12.9 \pm 4.5\times 10^{-10}, n_s \approx 1 - 7\alpha_3(M_P)/\pi = 0.958,$

Observed (Planck): $A = 21\pm 0.3\times 10^{-10}$, $n_s = 0.9587\pm 0.0056$

CPT-symmetry universe can solve lots of problems

(Proposed by Latham Boyle and Neil Turok)



How about perturbations?

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Evolution of Perturbations

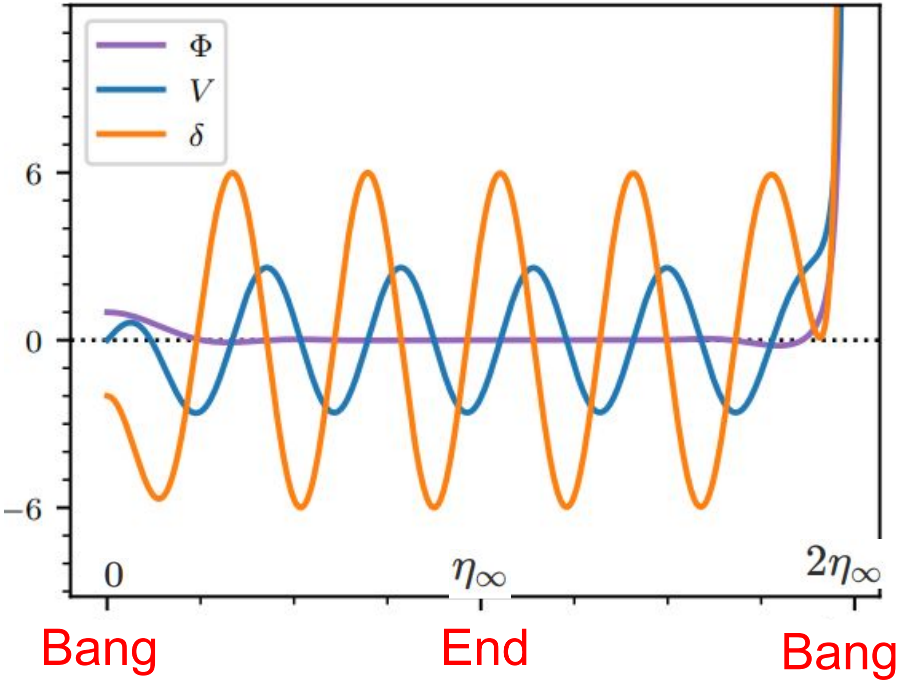
Lasenby et. al.[2104.02521]

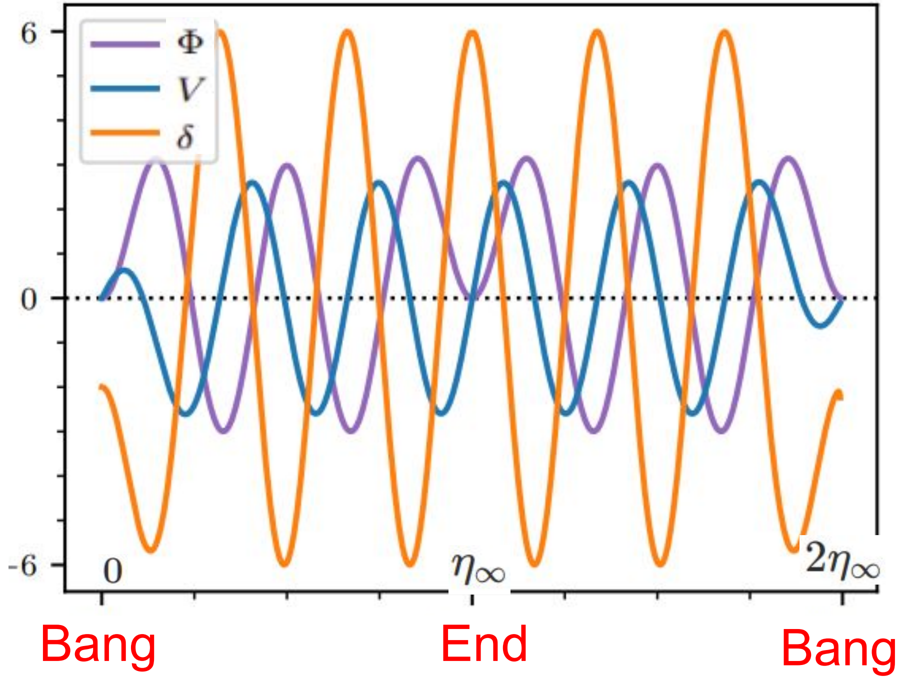
Φ : Newtonian potential
V : velocity perturbation of radiation
δ : density perturbation of radiation

$$\Phi(k,\eta) \propto \cos\theta(k,\eta)$$

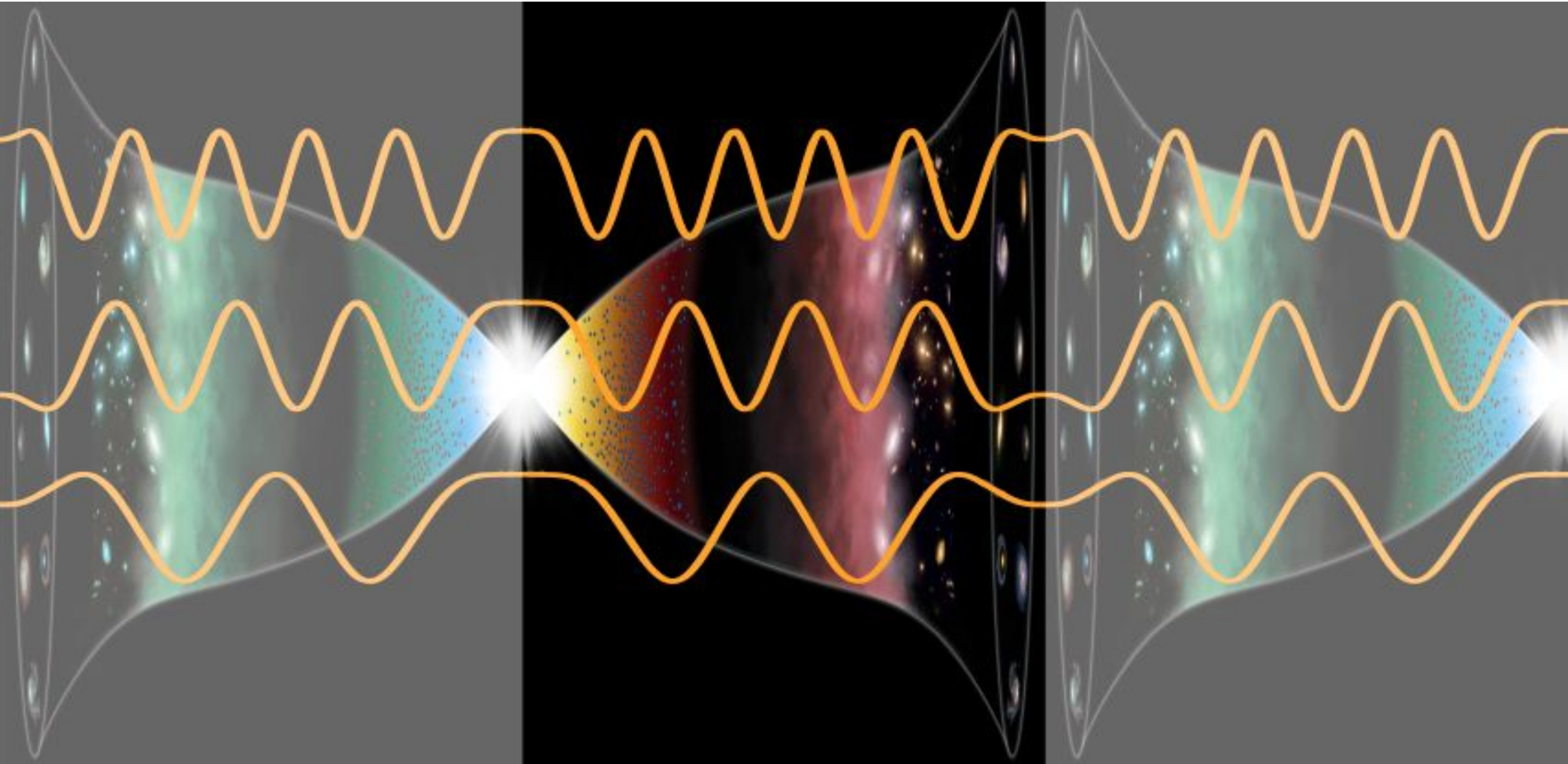
$$\boxed{\rightarrow \theta(k,\eta=\eta_\infty) = m\frac{\pi}{2}, m \in \mathbb{N}}$$

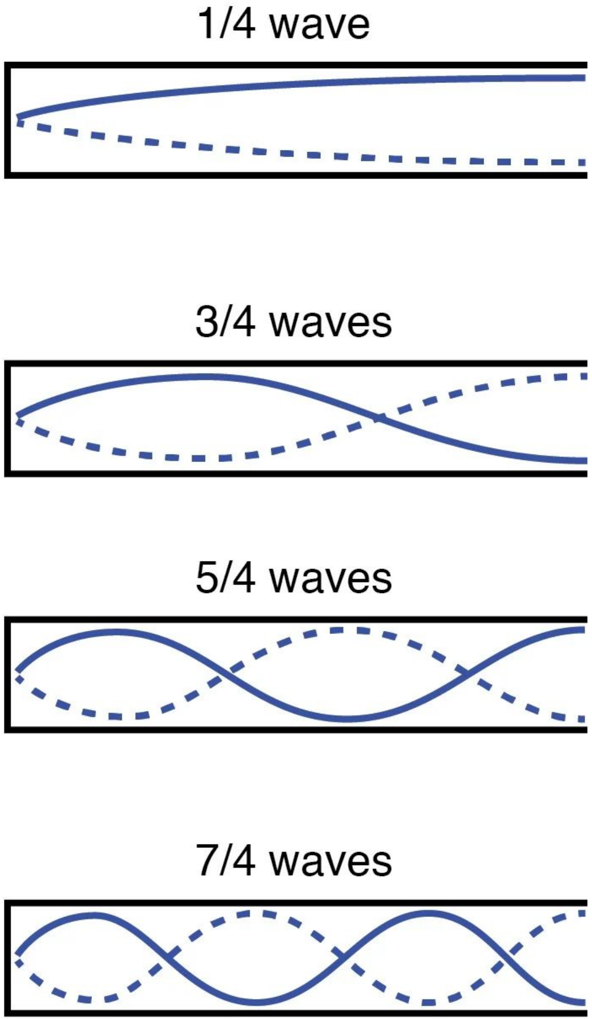
Perturbations also have to be symmetric!

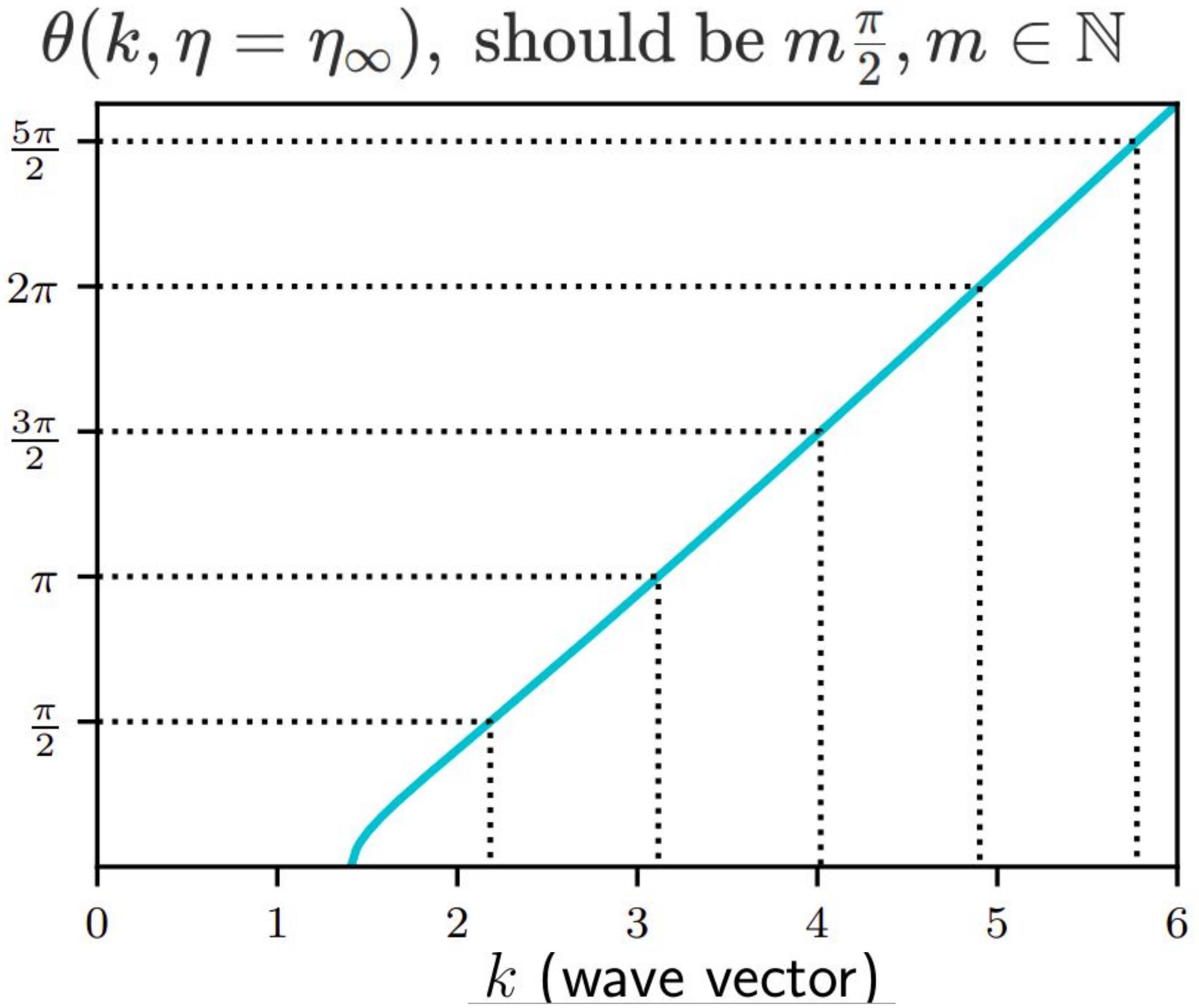




Perturbations should also be CPT-symmetric




1/4 wave
3/4 waves
5/4 waves
7/4 waves

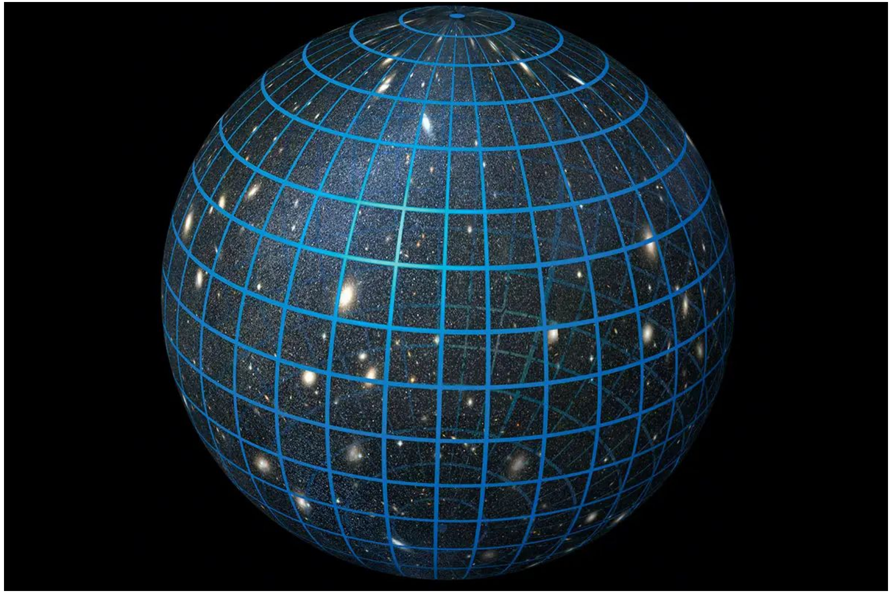

$\theta(k, \eta = \eta_\infty)$, should be $m\frac{\pi}{2}, m \in \mathbb{N}$
$\frac{5\pi}{2}$
2π
$\frac{3\pi}{2}$
π
$\frac{\pi}{2}$
0
1
2
3
4
5
6
k (wave vector)

Spatial boundary condition in closed universe

Closed universes have finite space ➡ an additional spatial boundary condition (in spherical harmonics)

➡ wave vectors have to be integers: $k = n \in [1, 2, 3, \ldots]$

Matching the two discrete k sets ➡ constraints curvature value.

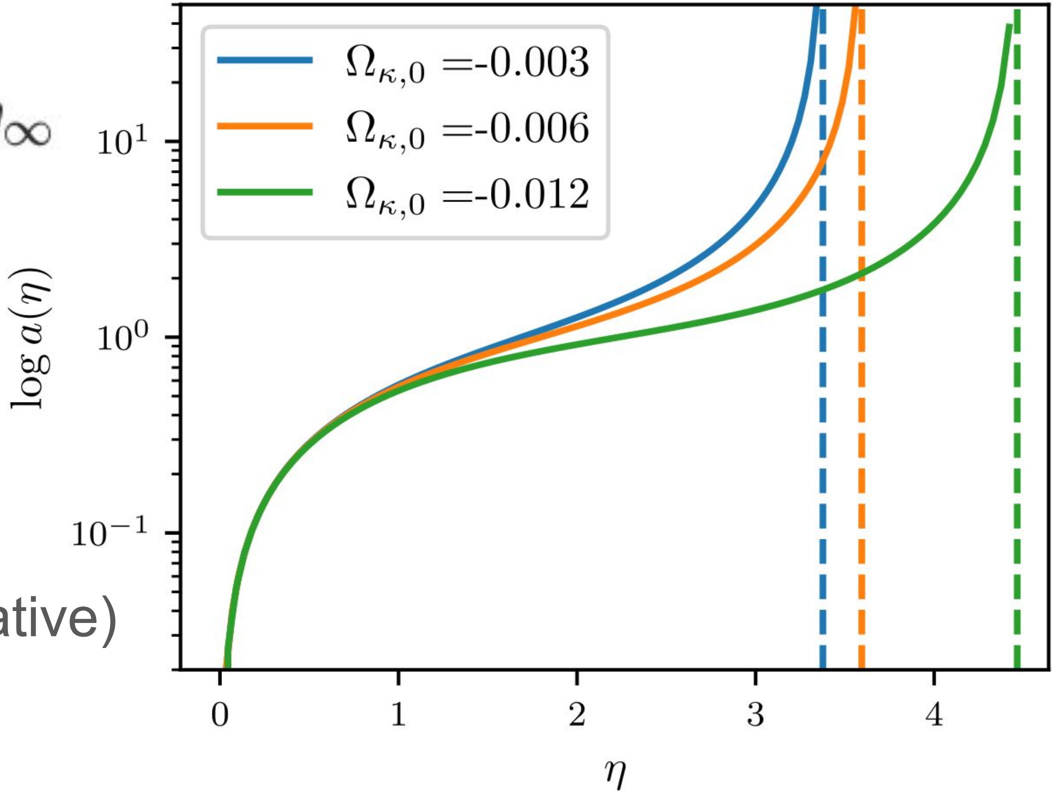


The end of the universe η_∞ depends on curvature

larger curvature $\longrightarrow$ larger η_∞

Since $\Delta k \propto 1/\eta_\infty$, larger η_∞ means smaller Δk.

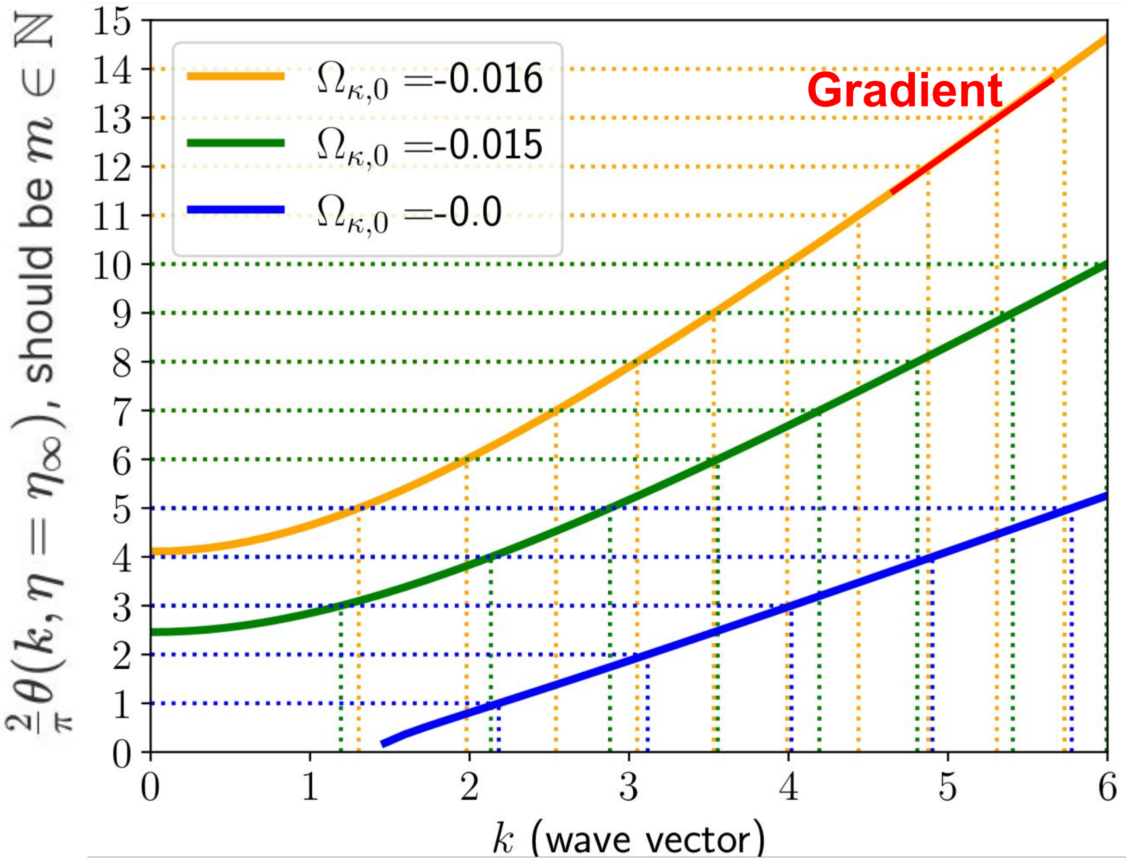
(larger curvature: $\Omega_{\kappa,0}$ more negative)



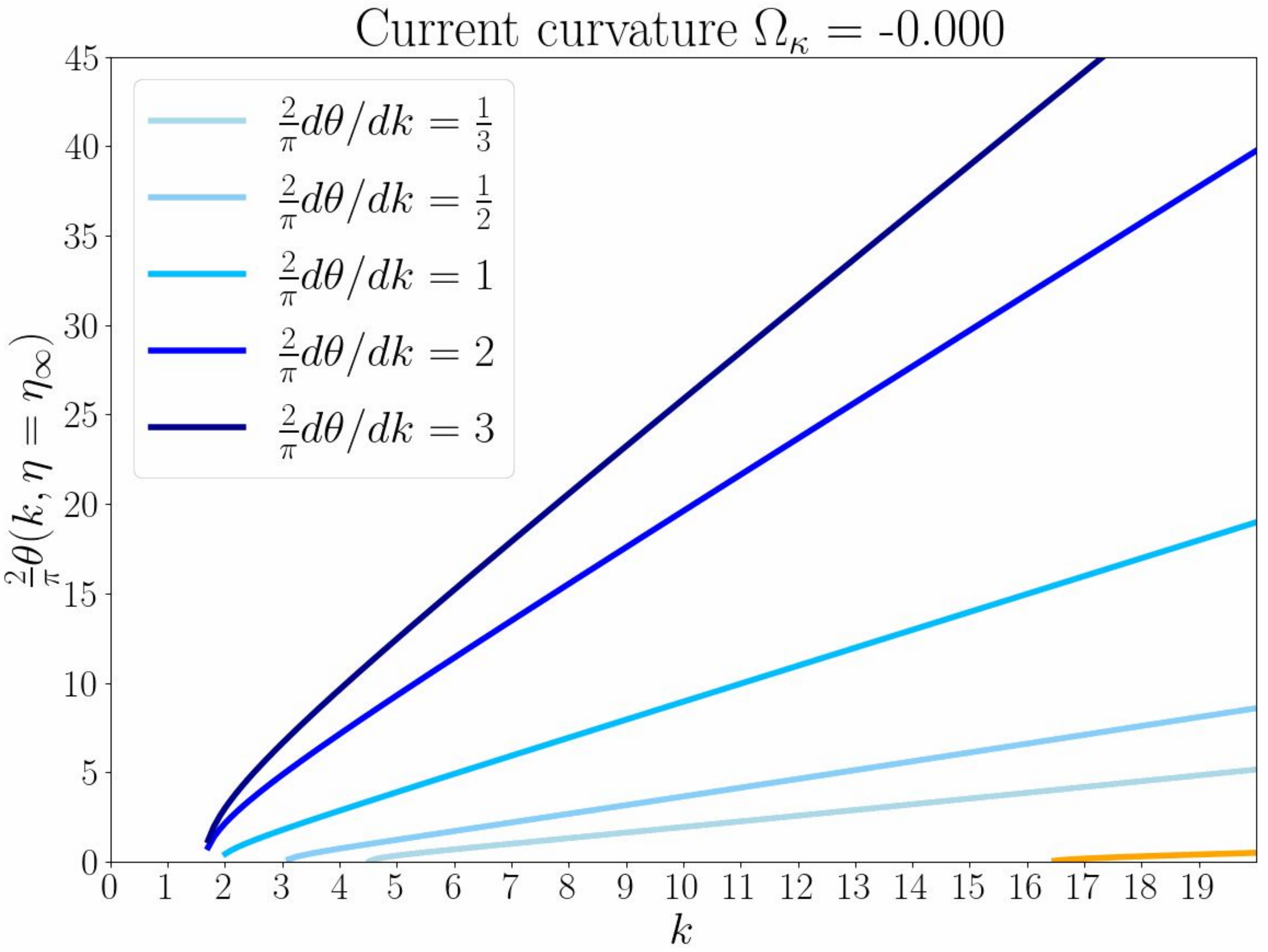
Idea:

Larger curvature has steeper $\theta(k, \eta = \eta_\infty)$
$\longrightarrow$ has discrete k with smaller spacing Δk.

Idea: Adjusting the curvature values, such that the discrete k become integers.



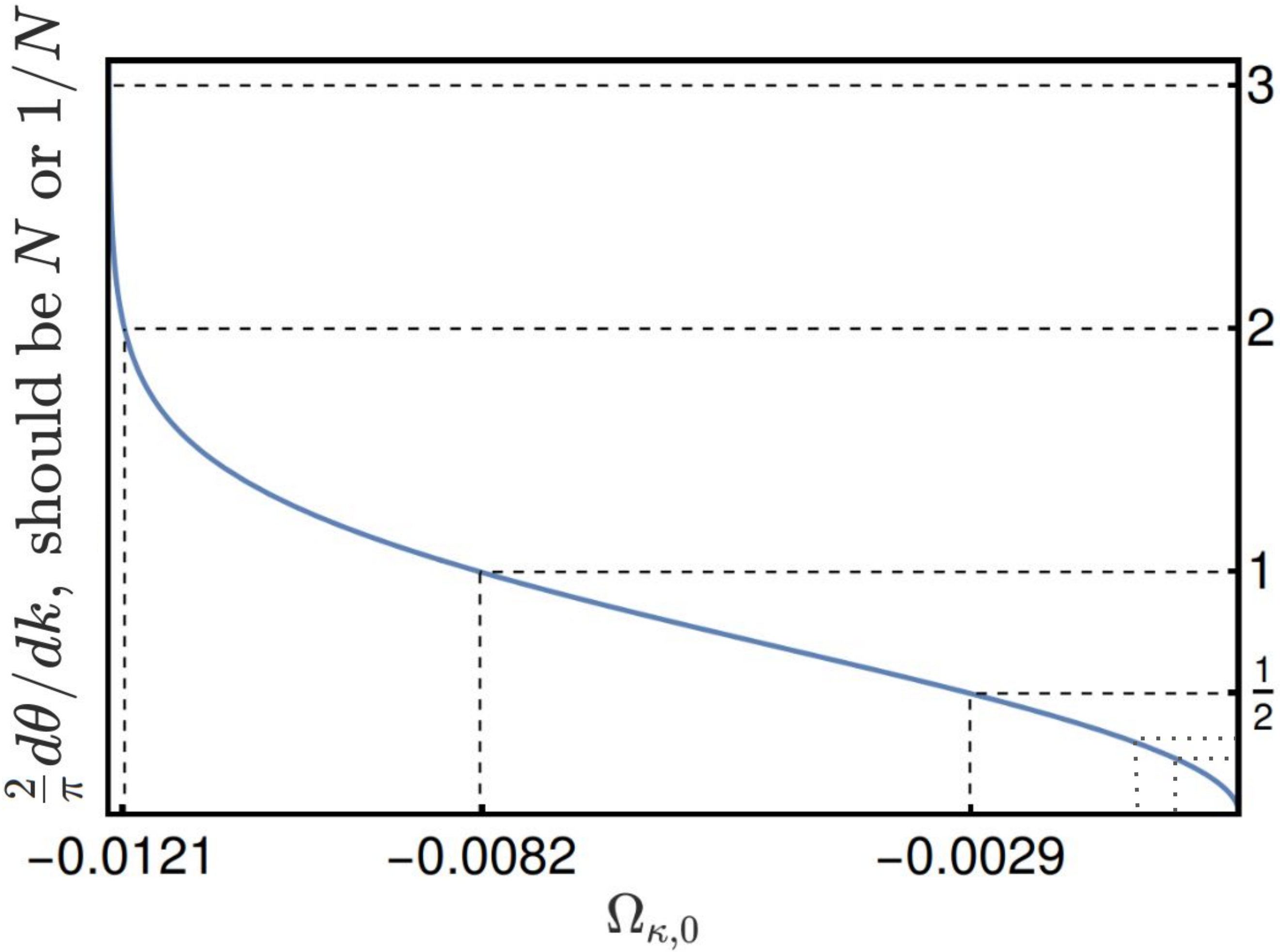
The gradient $\frac{2}{\pi}d\theta/dk$ should be N or $1/N$



Result

1. The gradient increase while increasing curvature.

2. It goes to infinity at $\Omega_{\kappa,0} \simeq -0.012$, curvature larger than this would cause big crunch.



Consider Real Cosmology

1. Solve Friedmann equation containing matter

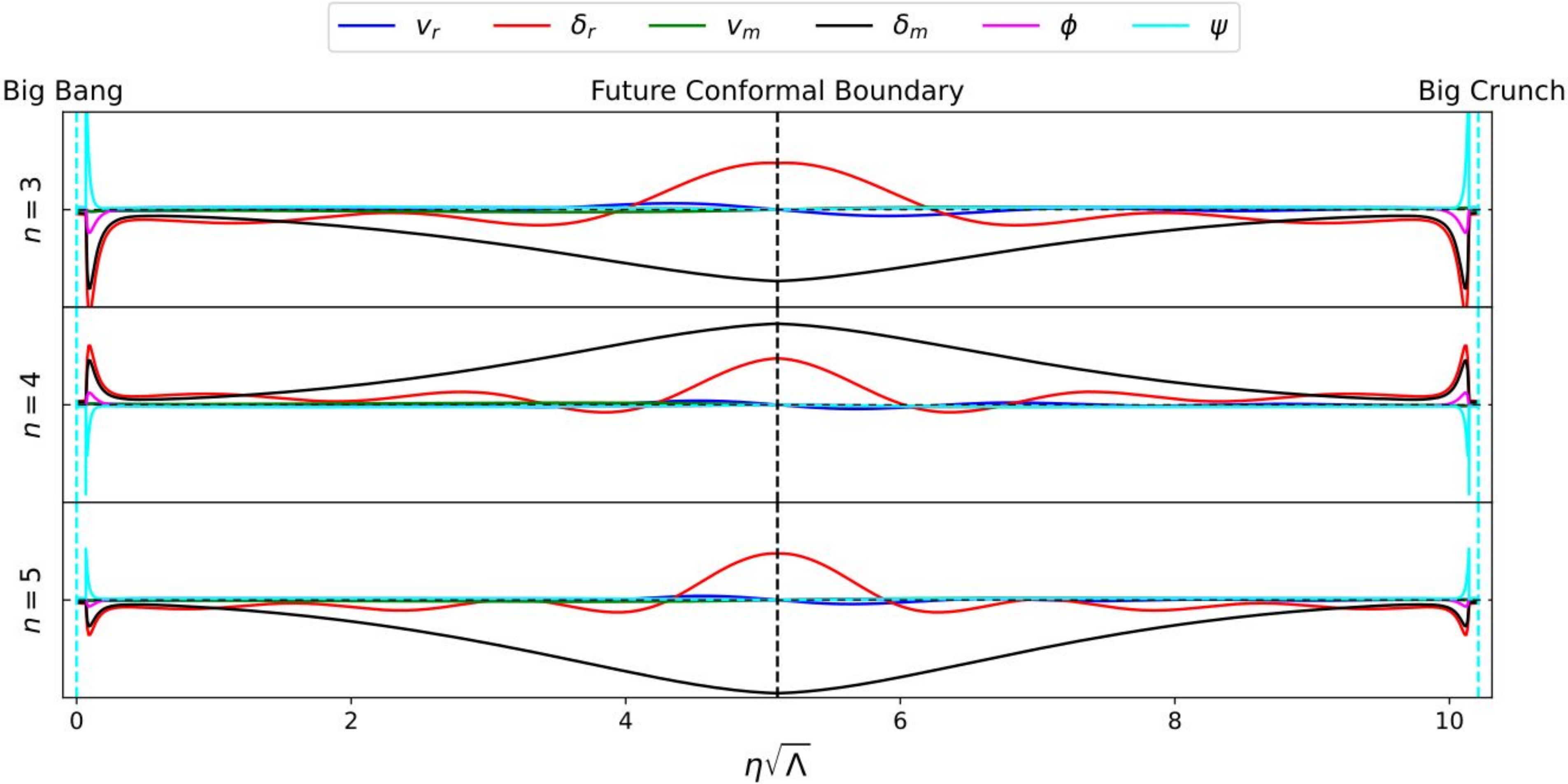
$$3\dot{a}^2 = \lambda(a^4 - 3\tilde{\kappa}a^2 + \tilde{m}a + \tilde{r}),$$

2. Transform to cosmological parameters

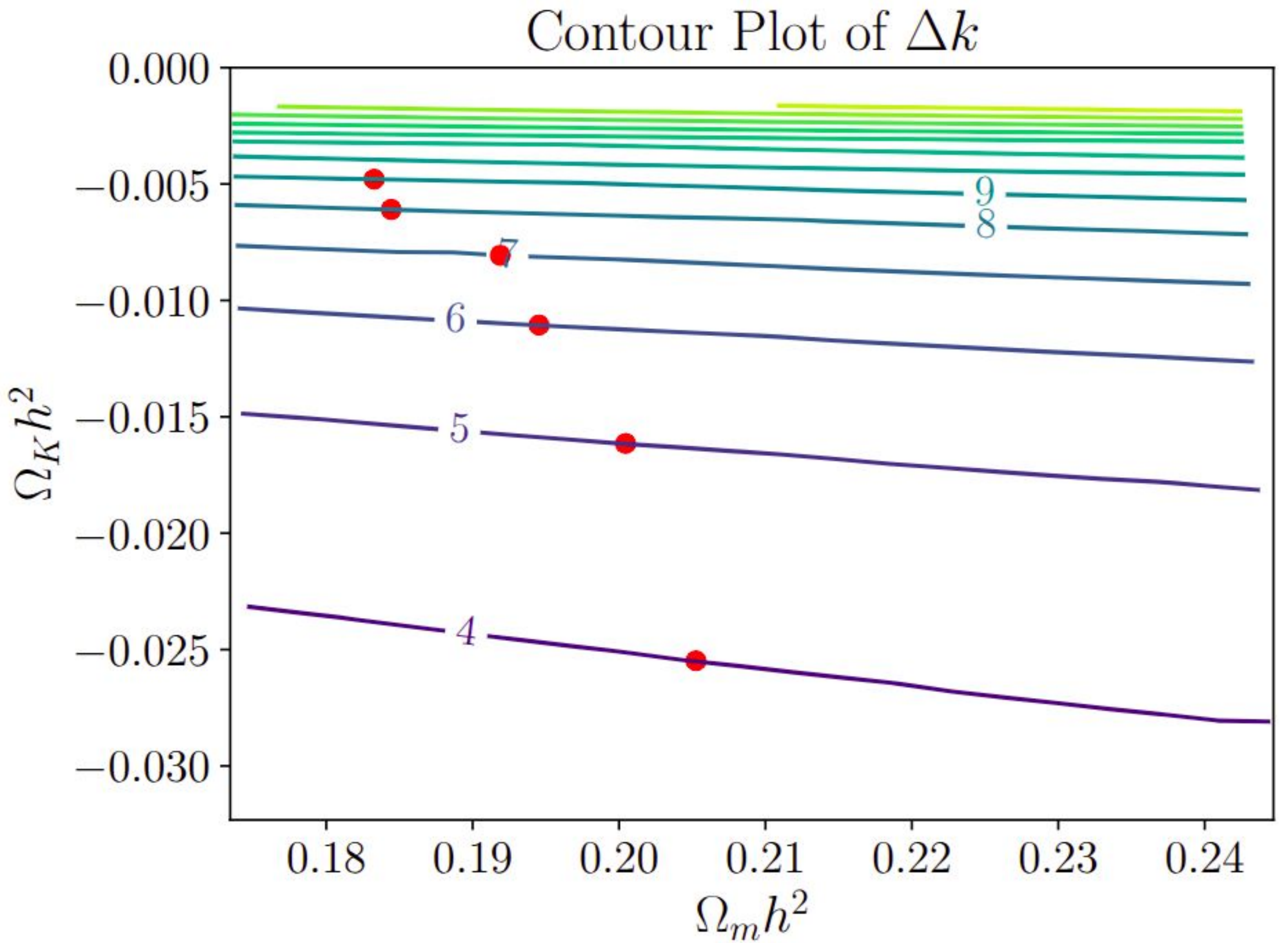
$$\Omega_m = \tilde{m}\Omega_\Lambda/a_0^3, \quad \Omega_K = -3\tilde{\kappa}\Omega_\Lambda/a_0^2, \quad \Omega_r = \tilde{r}\Omega_\Lambda/a_0^4,$$

3. Solve perturbation equations numerically (containing higher order Boltzmann terms) to get allowed wave vectors.

(Anti)-symmetric Perturbation solutions



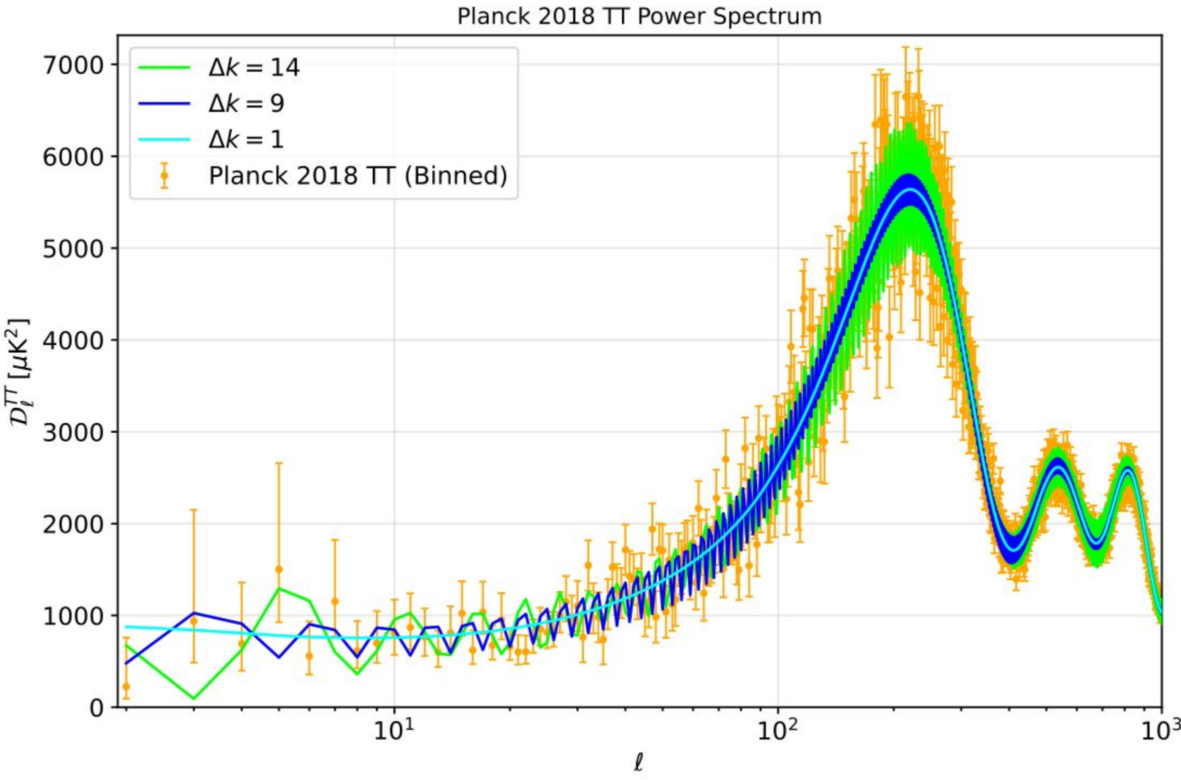
$(\Omega_K, \Omega_m, \Omega_b, h) = N \in \mathbb{N}$



Compare with CMB

Modify Boltzmann code CLASS to see how sparse wave vectors affect CMB power spectrum.

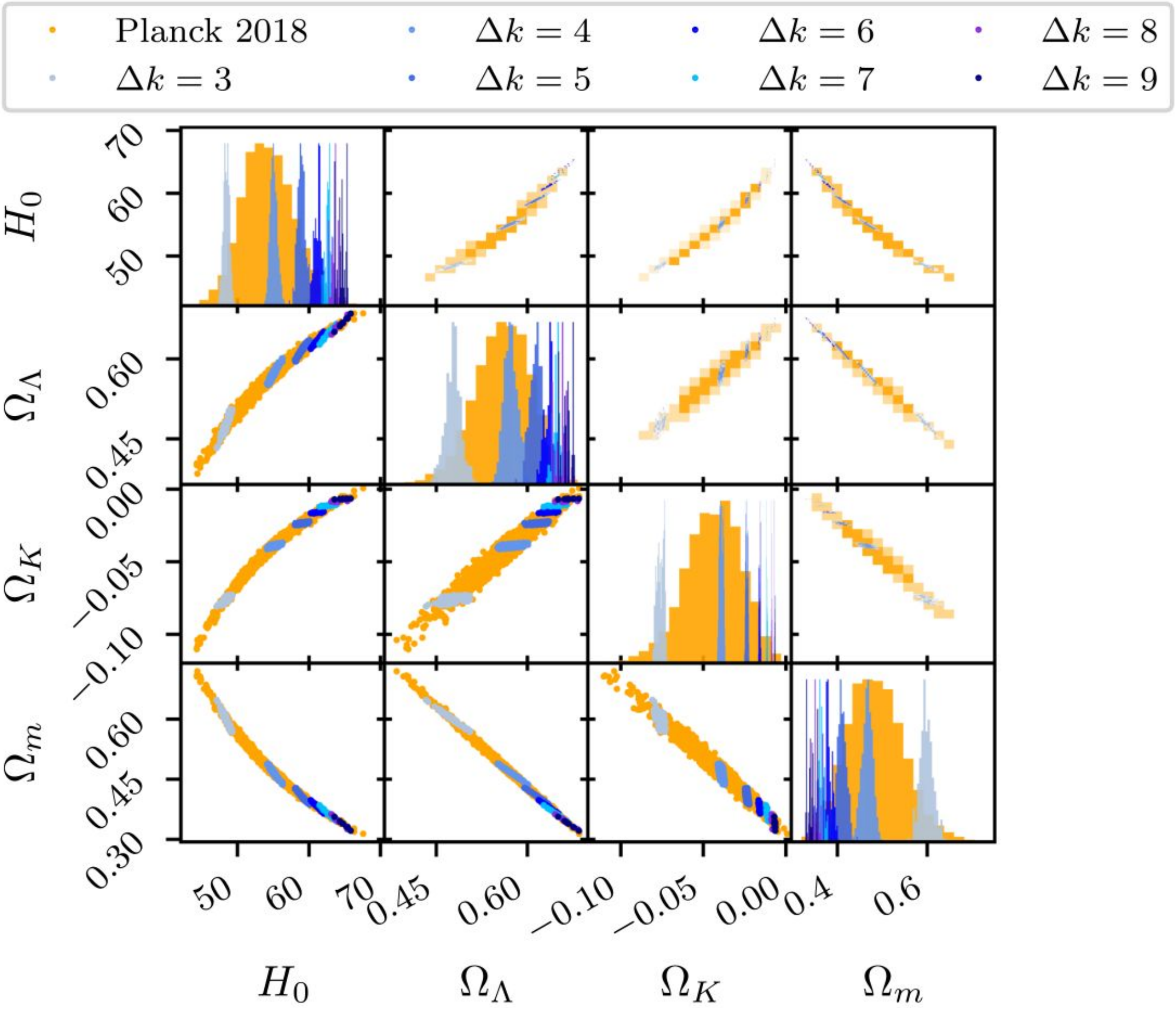
1. More fluctuations if N and curvature ($\tilde{\kappa}$) become large.
2. However, since $\Delta\tilde{k} = N\sqrt{\tilde{\kappa}}$, N and $\tilde{\kappa}$ are negatively correlated→never both large
3. The fluctuation behaviour could be ignored.

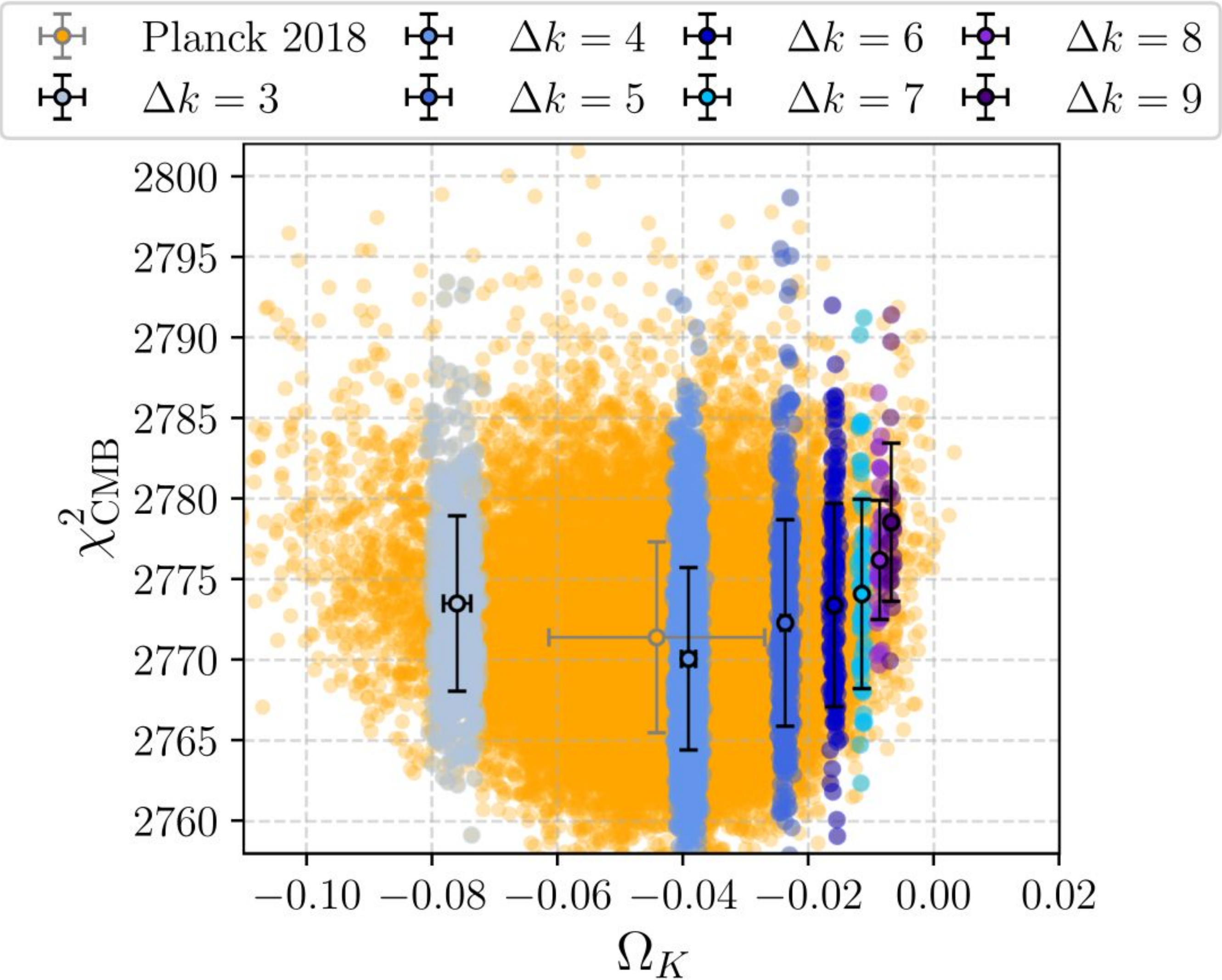


CMB constraint

Filter Planck MCMC data with the $\Delta k \in \mathbb{N}$ constraint.

Need joint analysis to break the degeneracy.





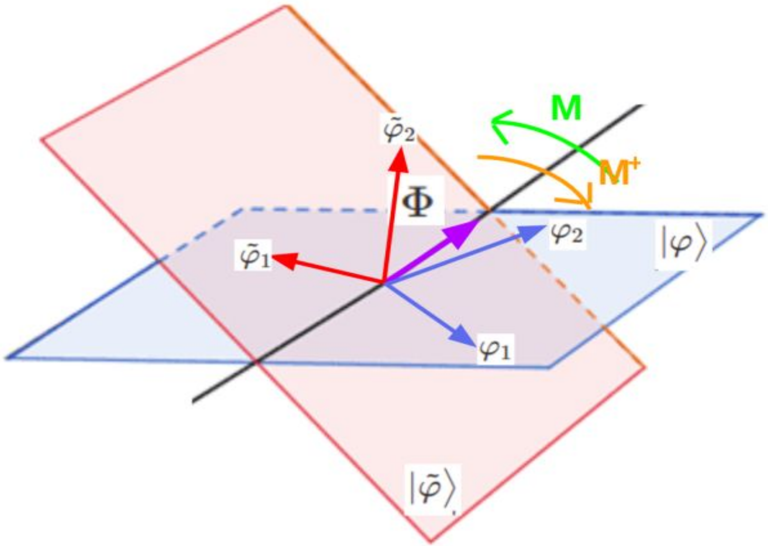
Planck 2018
$\Delta k = 3$
$\Delta k = 4$
$\Delta k = 5$
$\Delta k = 6$
$\Delta k = 7$
$\Delta k = 8$
$\Delta k = 9$
χ^2_{CMB}
2800
2795
2790
2785
2780
2775
2770
2765
2760
−0.10
−0.08
−0.06
−0.04
−0.02
0.00
0.02
Ω_K

How about low-k modes?

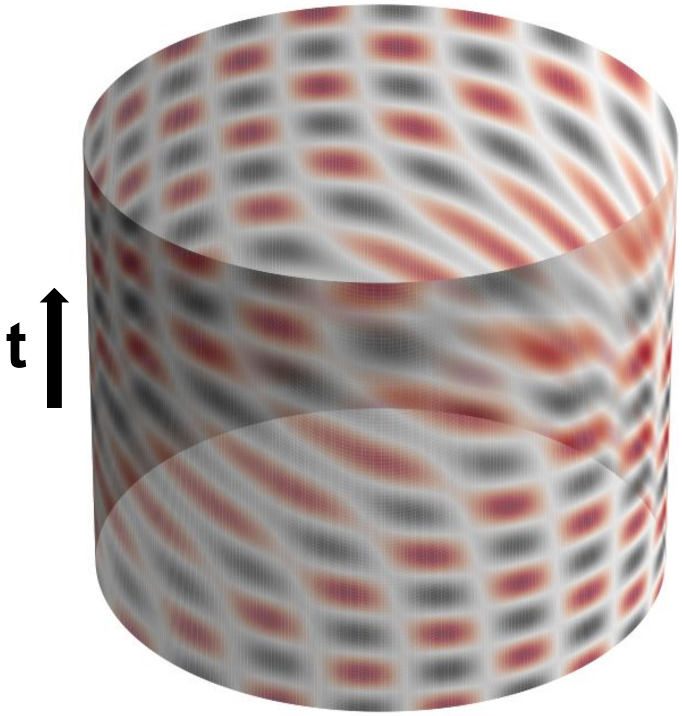
1. So far we only consider large k modes, where allowed wave vectors become equally spaced.
2. The valid solutions should satisfy both spatial and temporal BC →could be constructed by linear combination of both bases:

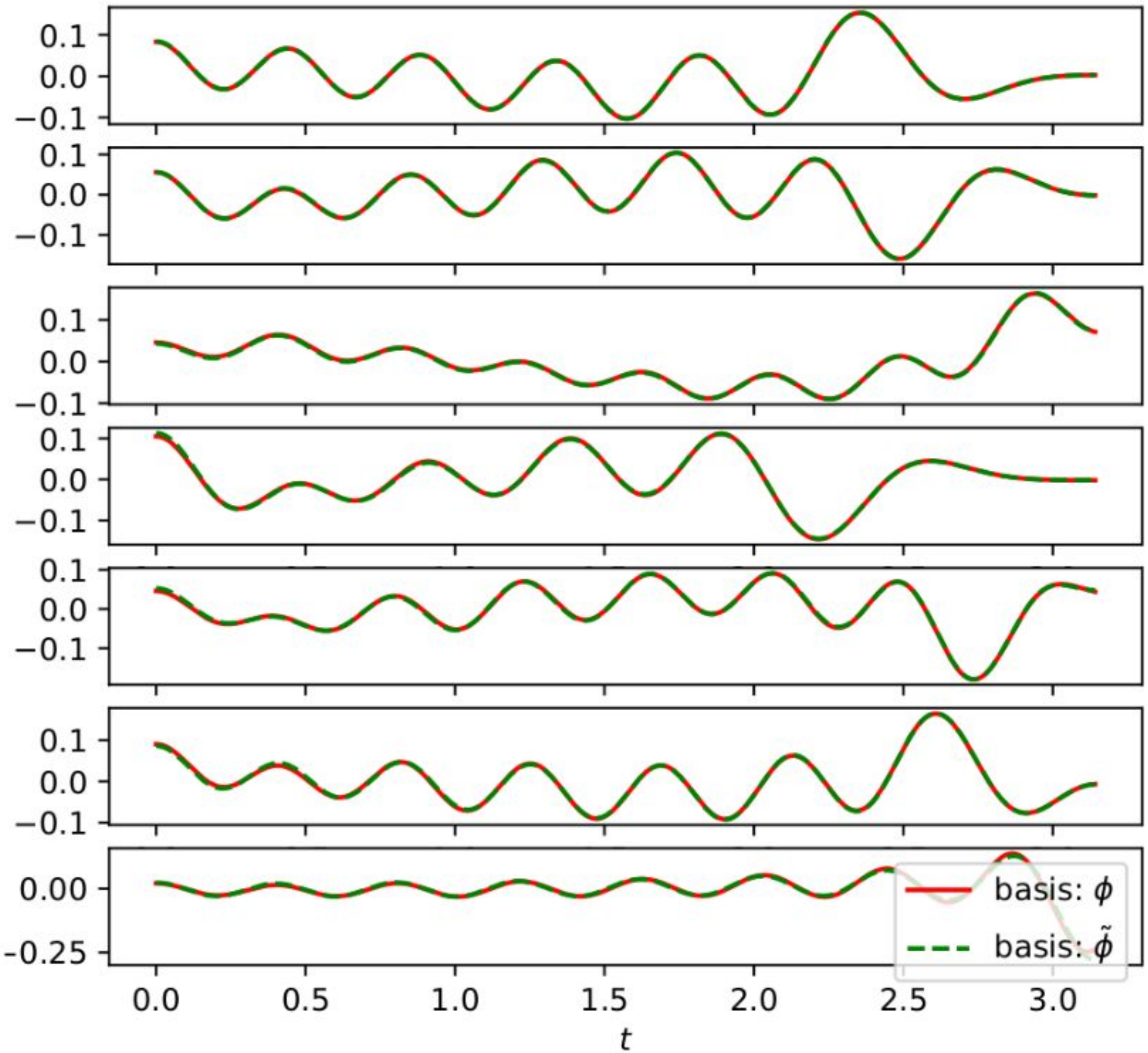
$$\Phi(x,t) = \sum_{n=1}^{\infty} c_n \phi_n = \sum_{m=1}^{\infty} \tilde{c}_m \tilde{\phi}_m$$

3. The linear combination coefficients are eigenvectors of mixing matrix M with eigenvalue = 1.



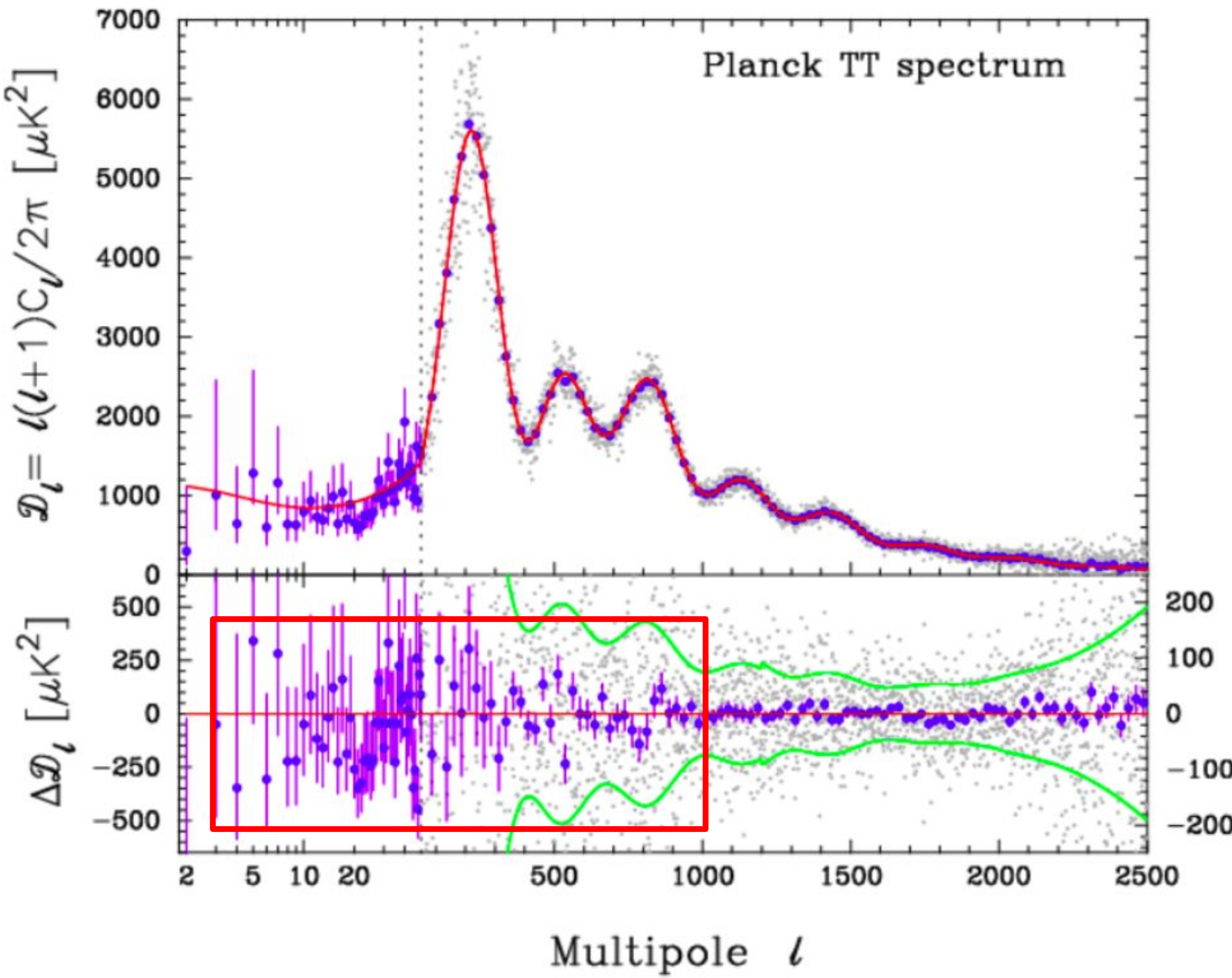
Result





Future work

Extend to cosmological perturbations. See whether could explain CMB residuals



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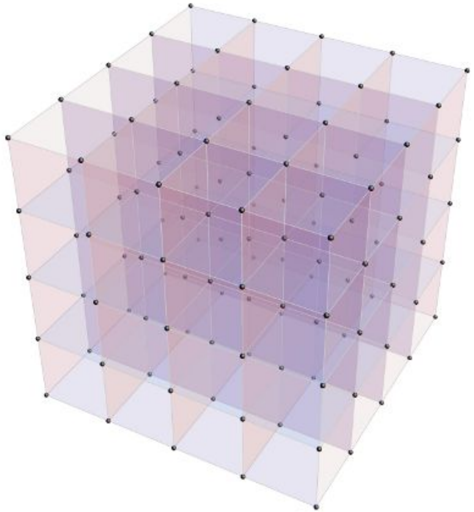
CPT-symmetric Kähler-Dirac fermions



Question: is CPT-symmetric preserved also in quantum scale?

Kähler-Dirac fermions:

(1) Written in Differential forms.

(2) Could solve fermion doubling problem.

Differential formulation: $(K - m)\Phi = 0, \qquad K = d - d^{\dagger}, \quad \Phi = (\phi, \phi_{\mu}, \phi_{\mu\nu}, \dots)$

Introduction: conventional Dirac equation

$$(i\gamma^\mu \partial_\mu - m)\psi = 0$$

1. Describe how fermions evolve.
2. First order equation – similar to Schrodinger equation – predict anti-particles
3. Could be reduced to second order equation (Klein-Gordon equation):

$$(i\gamma^\mu \partial_\mu - m)(i\gamma^\mu \partial_\mu + m)\psi = -(\partial^\mu \partial_\mu + m^2)\psi = 0$$

Problem: fermion doubling problem, only valid in flat (Minkowski) spacetime

$$(i\gamma^\mu D_\mu - m)\psi = 0, \qquad D_\mu = \partial_\mu + w_\mu$$

Is there any other way to write down Dirac equation?

Introduction: Kähler-Dirac equation

Original formulation: $(K - m)\Phi = 0, \qquad K = d - d^\dagger \;, \quad \Phi = (\phi, \phi_\mu, \phi_{\mu\nu}, \dots)$

Spinor formulation: $(i\gamma^\mu \partial_\mu - m)\Psi = 0, D_\mu \Psi = \partial_\mu \Psi + [w_\mu, \Psi], \; \Psi = \sum_{p=0}^{4} \frac{1}{p!} \gamma^{\mu_1 \dots \mu_p} \phi_{\mu_1 \dots \mu_p}$

4x4 spinor (4 Dirac or 8 Weyl): $\Psi_{4\times 4} = \left[\psi_1 \psi_2 \psi_3 \psi_4\right], \psi_i = \begin{pmatrix} \varphi \\ \chi \end{pmatrix}$

Written in differential forms – easy be extended to lattice & curved geometry.

1. Important application: Lattice field theory, numerical simulation fermions – treat space and time equally – **Euclidean space:**

$$ds^2 = d\tau^2 + dx^2 + dy^2 + dz^2, \qquad \tau = t/i$$

2. Possible application: curved spacetime, ex: early universe, near BH or boson stars – **Lorentzian space**:

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$$

Additional γ^0 in Lagrangian

Different observers should see the same physics – scalar should be Lorentz invariant:

1. Conventional Dirac 4x1 spinor:
   Lorentz transform: $\psi' = U_\Lambda \psi, U_\Lambda^\dagger = \gamma^0 U_\Lambda^{-1} \gamma^0$
   Scalar should be Lorentz invariant: $\bar{\psi}'\psi' = \bar{\psi}\psi \rightarrow \bar{\psi} = \psi^\dagger \gamma^0$
   Lagrangian: $\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi = \psi^\dagger \gamma^0 (i\gamma^\mu \partial_\mu - m)\psi$

2. Kähler-Dirac 4x4 spinor:
   Lorentz transform: $\Psi' = U_\Lambda \Psi \underline{U_\Lambda^{-1}}, U_\Lambda^\dagger = \gamma^0 U_\Lambda^{-1} \gamma^0,$
   Scalar should be Lorentz invariant: $\bar{\Psi}'\Psi' = \bar{\Psi}\Psi \rightarrow \bar{\Psi} = \underline{\gamma^0} \Psi^\dagger \gamma^0$
   Lagrangian: $\mathcal{L} = \mathrm{Tr}[\bar{\Psi}(i\gamma^\mu \partial_\mu - m)\Psi] = \mathrm{Tr}[\underline{\gamma^0} \Psi^\dagger \gamma^0 (i\gamma^\mu \partial_\mu - m)\Psi]$

Quantize Kähler-Dirac Fermions

1. Diagonalize the extra γ^0:

   $\gamma^0 = \Omega\Lambda\Omega^T$, with $\Lambda = \mathrm{diag}(1, 1, -1, -1)$ and $\Omega = (\Omega^T)^{-1} \in O(4)$.

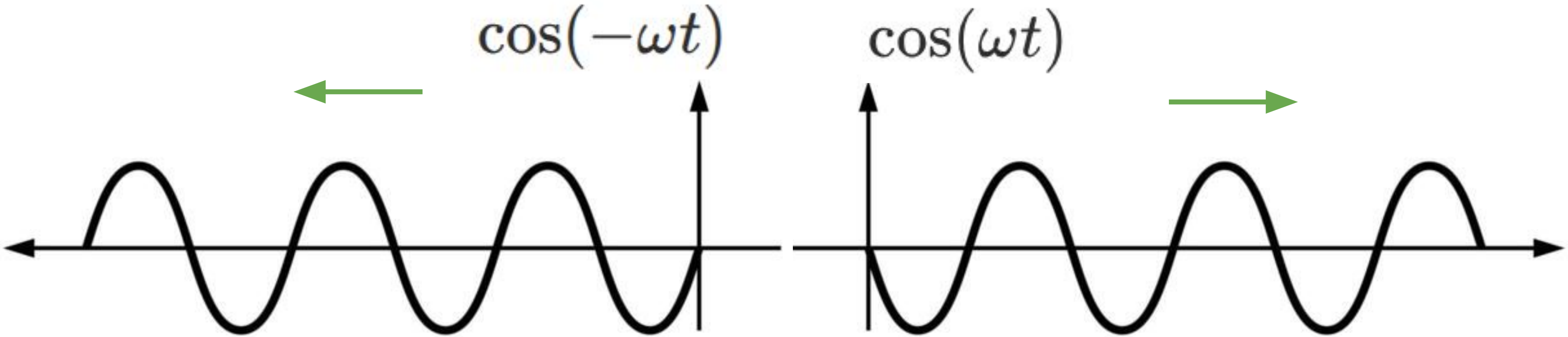
   In terms of $\Psi' \equiv \Psi\Omega$, the action is:

   $$S_{KD} = \int d^4x \, \mathrm{Tr}[\Lambda\Psi'^{\dagger}\gamma^0(i\partial\!\!\!/ - m)\Psi'].$$

2. The fields can be separated into first two and second two columns, which have positive and negative actions (Hamiltonian, or energy, or norm), respectively!

Positive/Negative energy (frequency)

Particles with positive/negative energy travel forward/backward in time.



KD-Majorana condition

Q: What are particles with negative E? Do they couple with the one with positive E?

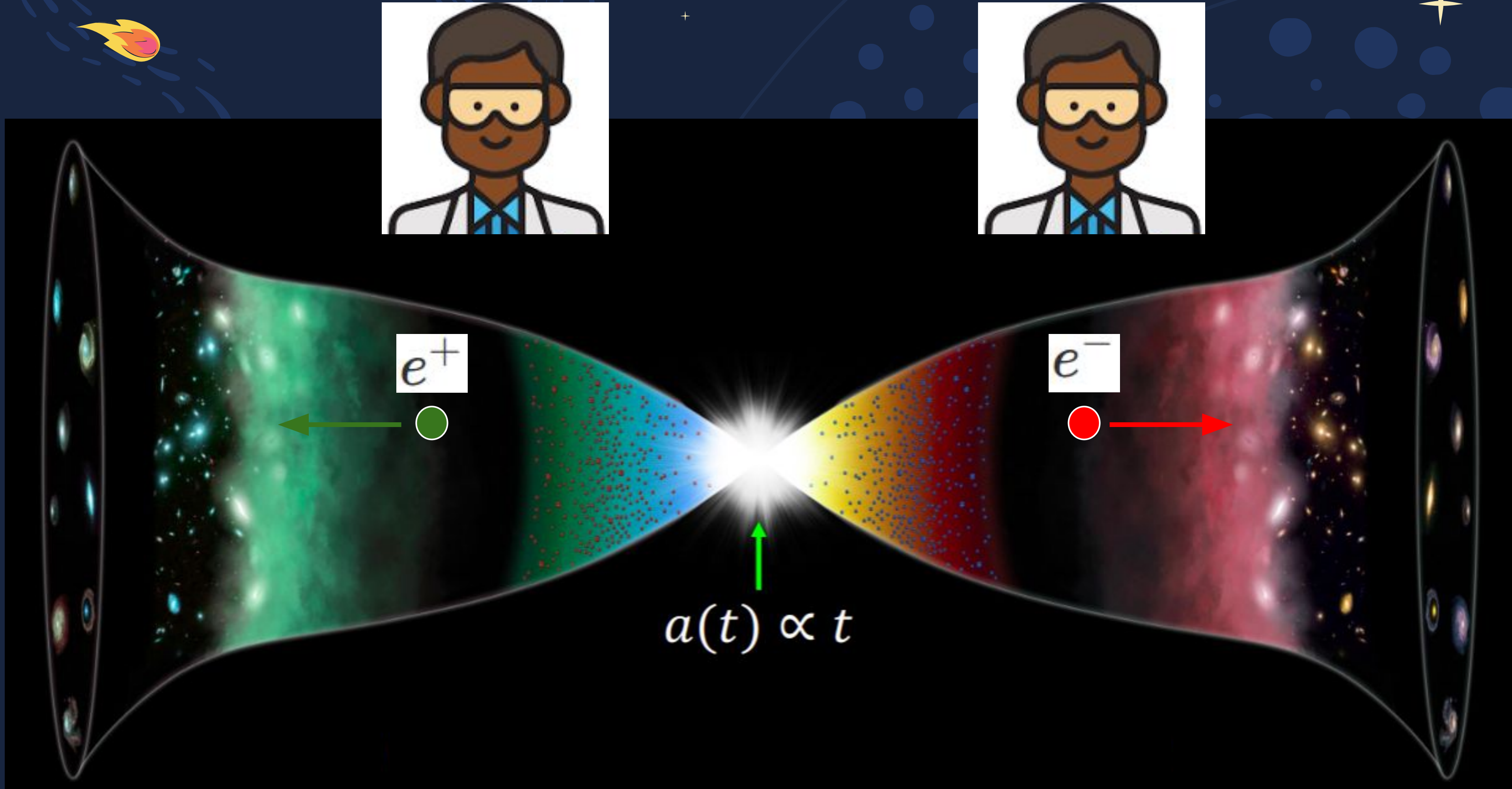
Kähler-Dirac complex conjugate: $\Psi'^{c} = (i\gamma^2)\Psi'^{*}(i\gamma^2)$

$$\Psi' = (\psi_1, \psi_2, 0, 0) \qquad \Psi'^{c} = (0, 0, -\psi_2^c, \psi_1^c)$$

KD-Majorana condition: $\Psi'^{c} = \Psi'$

$$\Psi' = (\psi_1, \psi_2, -\psi_2^c, \psi_1^c).$$

Particles/anti-particles in universe/anti-universe are totally CPT-symmetric!



e^+
e^-
$a(t) \propto t$

Relation to the standard model

1. Pati-Salam unification: all fermions are contained in 4 copies of KD: $\Psi'_{A=0,\ldots,3}$

   Leptons: $\Psi'_0 = \chi_0 + \chi_0^c$, where $\chi_0 = \begin{pmatrix} \nu_L & e_L & 0 & 0 \\ \nu_R & e_R & 0 & 0 \end{pmatrix}$,

   3 colors of quarks: $\Psi'_i = \chi_i + \chi_i^c$, where $\chi_i = \begin{pmatrix} u_L^i & d_L^i & 0 & 0 \\ u_R^i & d_R^i & 0 & 0 \end{pmatrix}$,

2. SM Lagrangian:

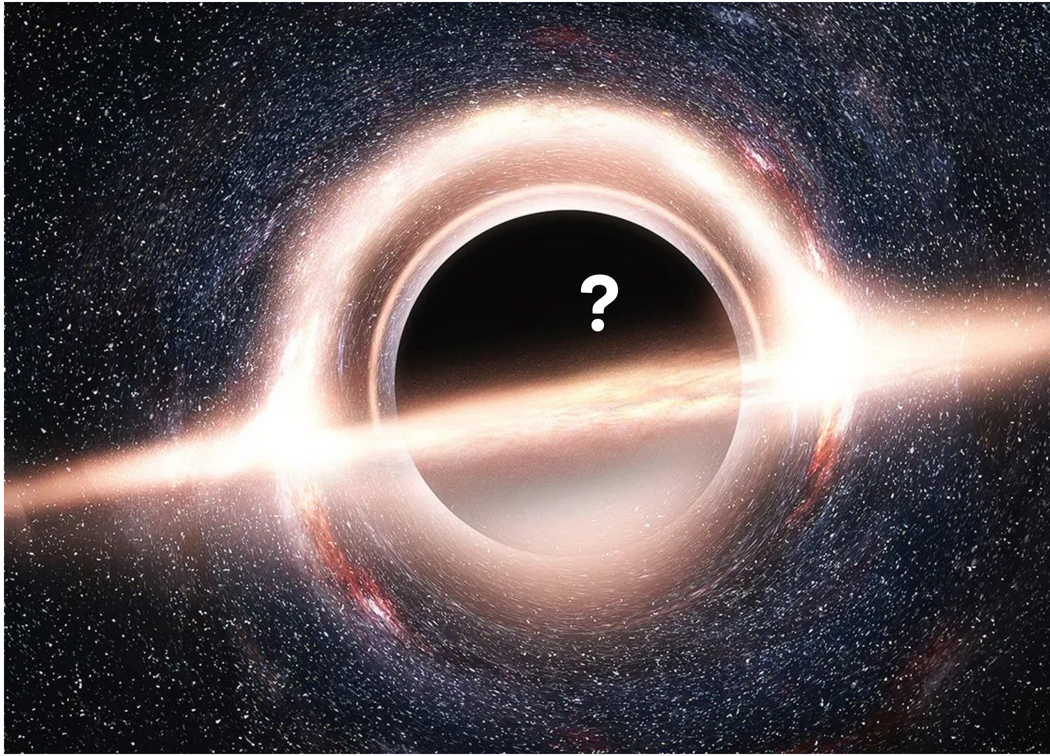
   Kinetic terms: $\mathcal{L}_{kin} = \mathrm{Tr}[\Lambda \Psi'^{\dagger}_A \gamma^0 (i\partial\!\!\!/) \Psi'_A]$,

   Yukawa coupling terms: $\mathcal{L}_Y = \mathrm{Tr}[\mathcal{H}^T \widehat{\Psi}'^{i}_{A,L} \Psi'^{j}_{A,R} \mathcal{Y}^{ij}_A] + \text{h.c.}$

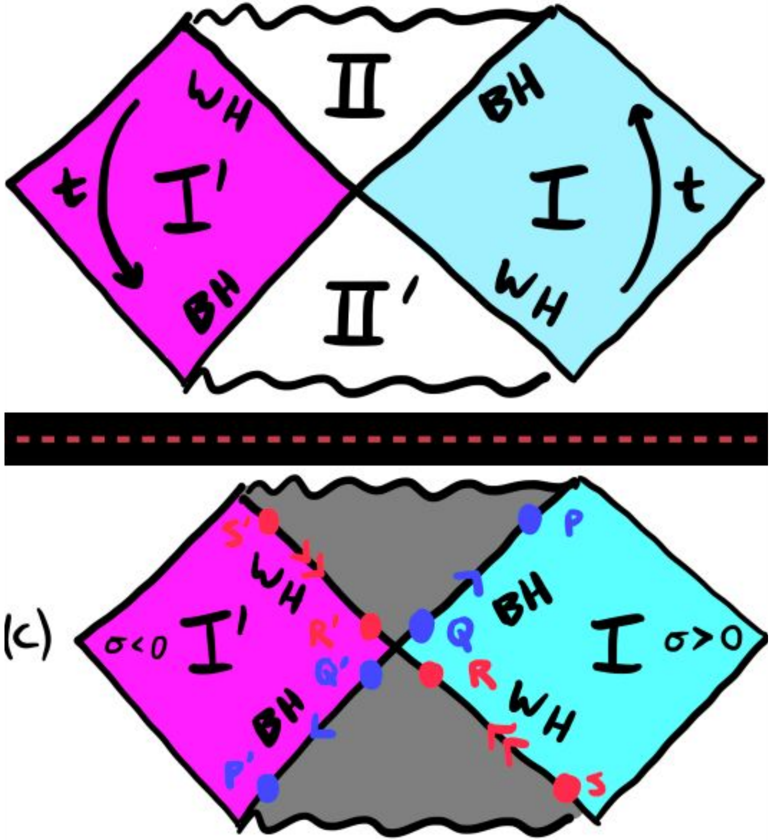
   Majorana mass term: $\mathcal{L}_M = \mathrm{Tr}[\widehat{\Psi}'^{i}_{A,R} \Psi'^{j}_{A,R} \mathcal{M}^{ij}_A]$

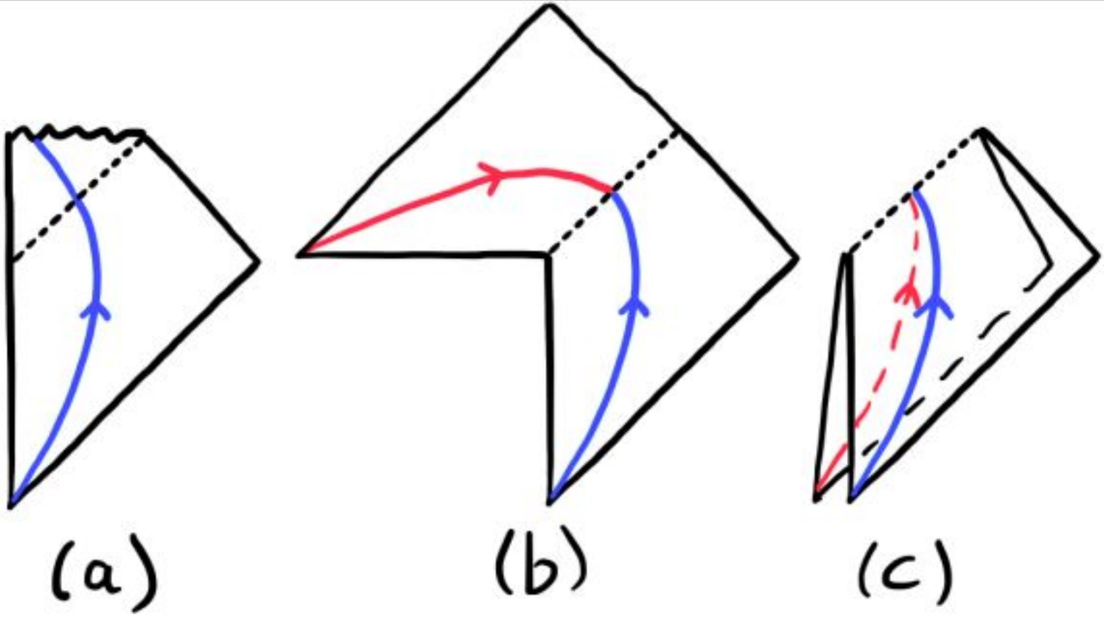
Black Hole Interior?

1. Hidden from Observation?
2. Curvature singularities
3. Cauchy horizons (breakdown of causality)
4. Information paradox
5. Does BH evaporate before inflating matter gets in?
6. Violates all global symmetries? (CPT? B-L?)



Black Hole Mirror





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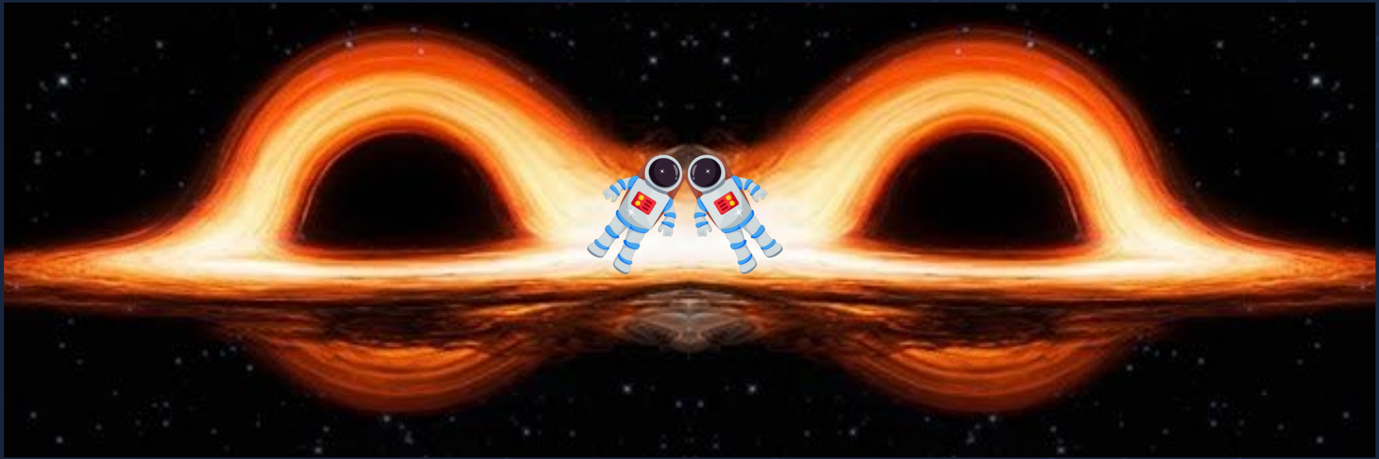
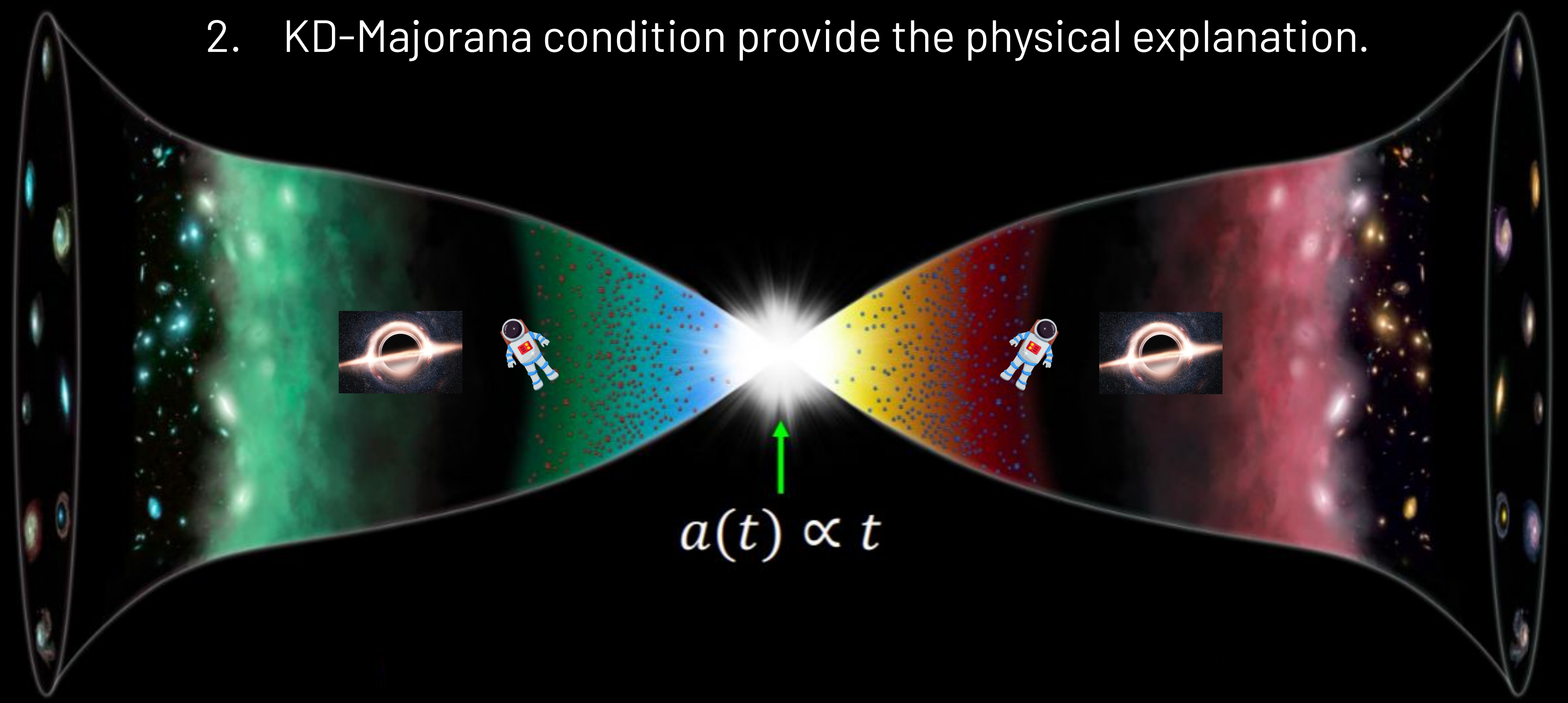


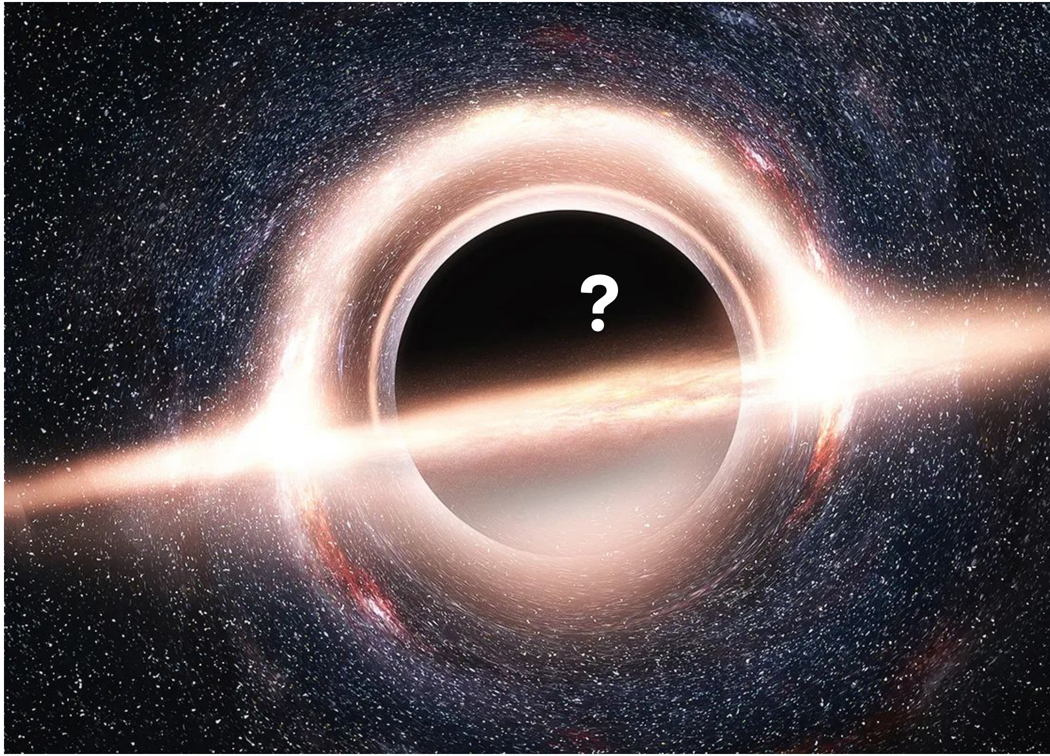
1. Hidden assumption: universe and anti-universe are totally symmetric, even in quantum scale.

2. KD-Majorana condition provide the physical explanation.



How BH mirror solve interior problems

1. BH has no interior – couldn't be observed, **no** Cauchy horizons, no Information paradox.
2. The metric is analytic and smooth across the horizon – **no** Curvature singularities
3. Particles & anti-particles annihilate at the horizon – all global symmetries preserved.



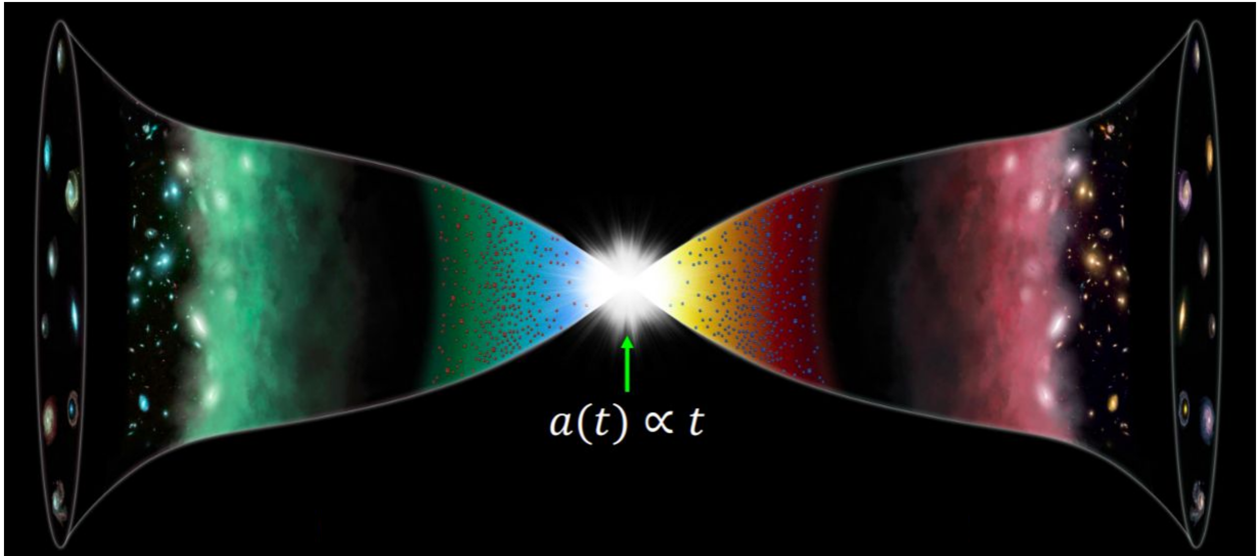
Theoretical explanation of Tenet?

Theoretical explanation of Tenet?

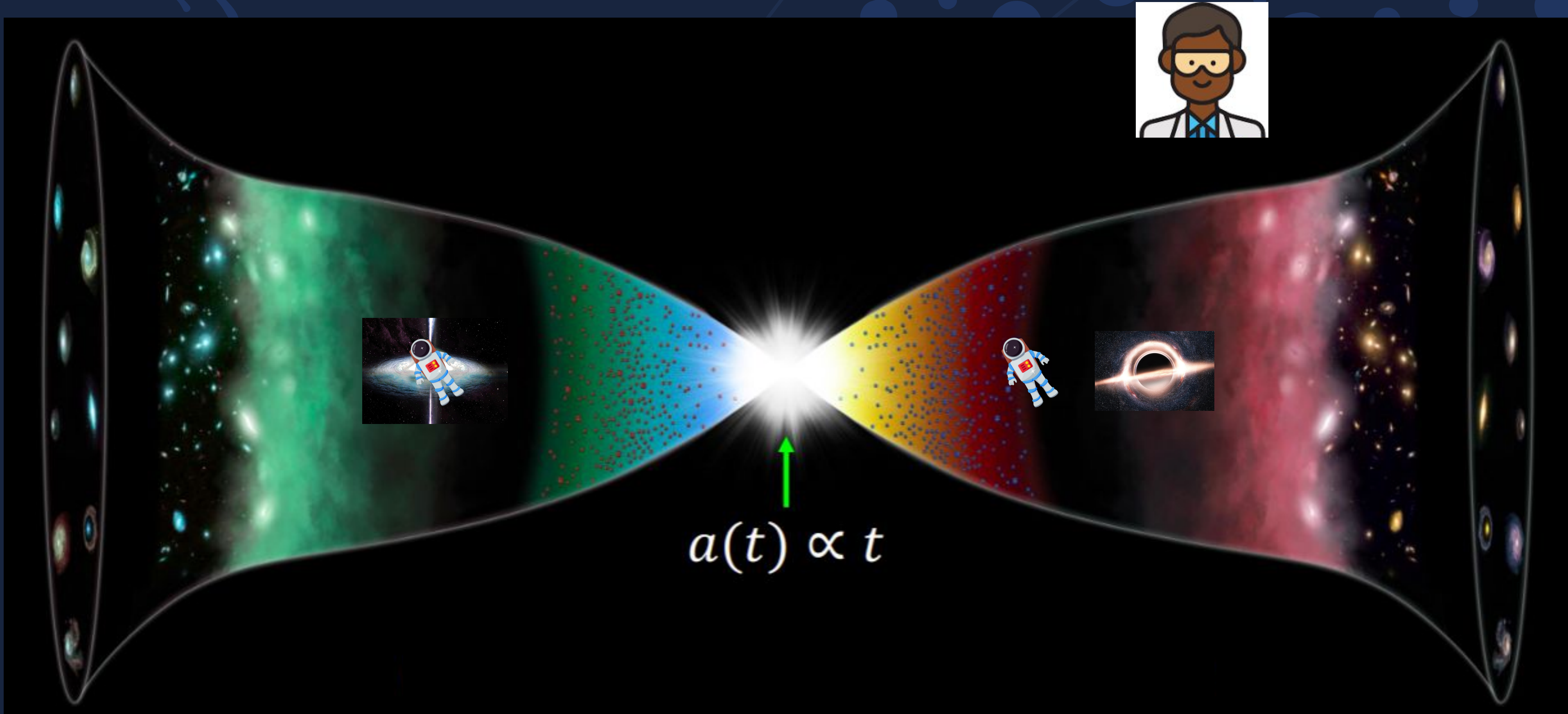
Q1: Is there exist an anti-universe where time goes backward?

Q2: Could we go to anti-universe?

Q3: Could we interact with particles in anti-universe?



BH mirror is the gate!



Theoretical description of Tenet?!

1. Same:
   (1) there is you and anti-you in the universe and anti-universe.
   (2) time go backward in anti-universe
   (3) the doors connecting the universe and anti-universe are black holes.

2. Difference:
   (1) You and anti you annihilate at the transition door.
   (2) If view in the coordinate of our universe: you will come out of a white hole and find that time goes backward. However, you cannot change anything but re-experience the past.

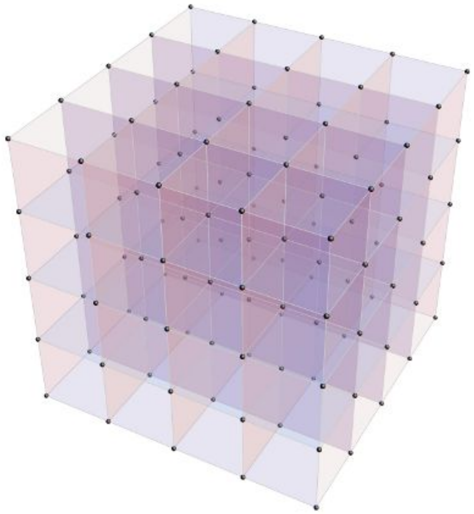
Future work for Kähler-Dirac Fermions

1. Lattice gauge theory: two sheeted spacetime provides an anomaly free solution.
2. Understand its topological properties through twistor theory

   (1) Why KD fields follow $Spin(3,1)_{\mathrm{Lorentz}} \times U(2,2)_{\mathrm{internal}}$ symmetry?

   (2) Could 4 copies naturally embedded in twistor space?

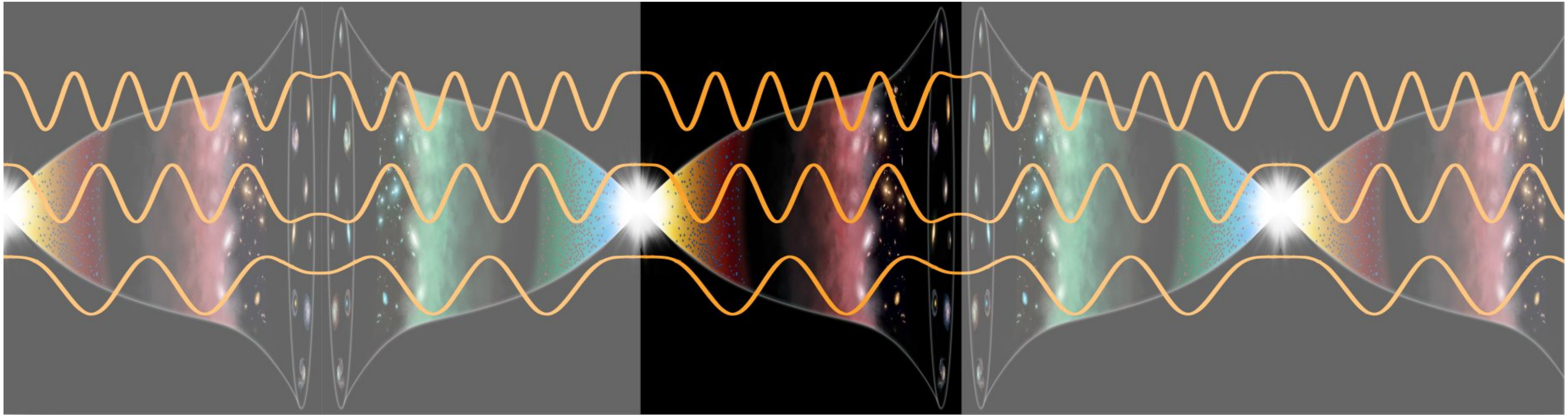
   (3) Why 3 generation of fermions?



Summary

Wei-Ning Deng (wnd22@cam.ac.uk)
PRD.110.103528, PRD.113.023546, arxiv 2511.11548

1. CPT-symmetric universe could solve lots of problems in cosmology and PP. ex: cosmological constant problem, DM, flatness & homogeneity problem.
2. Temporal BC puts constraint on spatial curvature values in closed universe
$$\Omega_K \in [-0.076, -0.039, -0.024, -0.016, -0.012, \ldots]$$
3. Kähler-Dirac Fermions provide physical explanation for CPT-symmetric condition in quantum scale.



Appendix

These fields can cancel the above 3 anomalies in coupling the SM to gravity

L. Boyle+NT, arXiv:2110.06258

Dim 1 scalars — fermions — gauge bosons — Dim 0 scalars

1. Vacuum energy $\propto n_{s,1} - 2n_F + 2n_A + 2n_{s,0}$
2. Conformal anomaly (Euler) $\propto n_{s,1} + \frac{11}{2} n_F + 62\, n_A - 28\, n_{s,0}$
3. Conformal anomaly (Weyl^2) $\propto n_{s,1} + 3\, n_F + 12\, n_A - 8\, n_{s,0}$

1) All three vanish iff $n_{s,1} = 0 \Longrightarrow$ no fundamental dimension one scalars (the Higgs must be composite)

2) Any two equations then give $n_F = 4n_A$ and $n_{s,0} = 3n_A$

3) For gauge group $SU3 \times SU2 \times U1$, *predict* n_F =48, *i.e.*, 3 fermion generations, each with a RH ν

Appendix

Solution of scale factor (in conformal time)

FLRW metric: $ds^2 = a^2(\eta)(-d\eta^2 + \gamma_{ij}dx^i dx^j)$

Friedmann equation: $\dot{a}^2 = \frac{1}{3}(r - 3\kappa a^2 + \lambda a^4)$

Solution (a Jacobi elliptic function):

$$\boxed{a(\eta) = \alpha\, \mathrm{sn}(\beta\eta,\, m),}$$

Where $\alpha = \sqrt{\frac{r(1+m)}{3\kappa}}, \quad \beta = \sqrt{\frac{\kappa}{(1+m)}}$

m is either roots of $\frac{(1+m)^2}{m} = \frac{(3\kappa)^2}{\lambda r}$

Solution of perturbations

Perturbation equation:

$$\ddot{\Phi} + 4aH\dot{\Phi} + \tfrac{1}{3}(k^2 + 4a^2\lambda - 12\kappa)\Phi = 0$$

Solution:

$$\Phi(a) = \frac{3\sqrt{a^4 + (\tilde{k}^2 - 2\tilde{\kappa})a^2 + 1}}{a^3\tilde{k}\sqrt{(\tilde{k}^2 - 2\tilde{\kappa})^2 - 4}} \sin\theta(\tilde{k}, \tilde{\kappa}, a),$$

$$\theta = \int_0^a \frac{\tilde{k}a'^2\sqrt{(\tilde{k}^2 - 2\tilde{\kappa})^2 - 4}}{\sqrt{1 + a'^4 - 2\tilde{\kappa}a'^2}\left(a'^4 + (\tilde{k}^2 - 2\tilde{\kappa})a'^2 + 1\right)} da'$$

Perturbation equations

$$(k_{\rm com}^2 - 3\kappa)\Phi = -\frac{1}{2}a^2\sum_i\left(\delta\rho_i - 3\frac{\dot{a}}{a}(\bar{\rho}_i + \bar{P}_i)v_i\right),$$

$$\dot{\Phi} = -\frac{\dot{a}}{a}\Psi - \frac{1}{2}a^2\sum_i(1+w_i)\rho_i v_i,$$

$$k_{\rm com}^2(\Phi - \Psi) = \frac{3}{2}a^2\sum_i(\bar{\rho}_i + \bar{P}_i)\sigma_i,$$

$$\dot{\delta}_i = (1+w_i)(3\dot{\Phi} + v_i k_{\rm com}^2),$$

$$\dot{v}_i = 3\frac{\dot{a}}{a}\left(w_i - \frac{1}{3}\right)v_i - \frac{w_i}{1+w_i}\left(\delta_i - \frac{2}{3}\left(1 - \frac{3\kappa}{k_{\rm com}^2}\right)\Pi_i\right)$$
$$-\Psi - \frac{a n_e \sigma_T}{k_{\rm com}^2}(\theta_b - \theta_r).$$

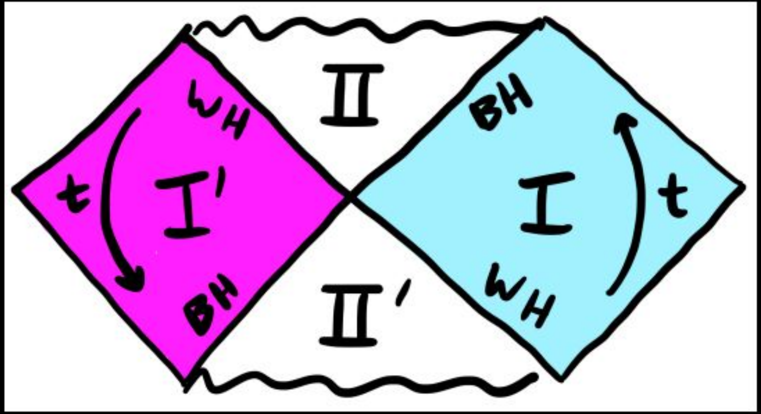
$$\dot{F}_{r\ell} = \frac{k_{\rm com}}{2\ell+1}[\ell F_{r(\ell-1)} - (\ell+1)F_{r(\ell+1)}] - n_e\sigma_T|a|F_{r\ell}.$$

Appendix

The Black Mirror Solution

$$ds^2 = -\frac{2m}{r(\sigma)}\left(\frac{\sigma}{4m}\right)^2 dt^2 + \frac{r(\sigma)}{2m}d\sigma^2 + r(\sigma)^2 d\Omega^2$$

$$r(\sigma) = 2m[1 + (\frac{\sigma}{4m})^2]$$



$$ds^2 = -\frac{2m}{r(\sigma)}\left(\frac{\sigma}{4m}\right)^2 dv_\pm^2 \pm \frac{\sigma}{2m}d\sigma\, dv_\pm + r(\sigma)^2 d\Omega^2$$

$$dv_\pm = dt \pm \frac{dr}{f(r)}$$

(c)